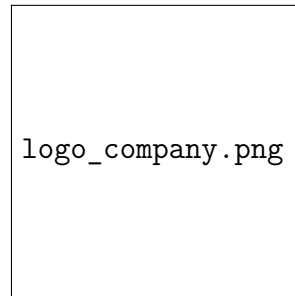
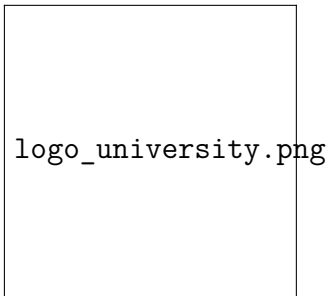


REPUBLIC OF TUNISIA
MINISTRY OF HIGHER EDUCATION AND SCIENTIFIC RESEARCH
[University / Institute Name]



FINAL-YEAR GRADUATION PROJECT THESIS
submitted in partial fulfilment of the requirements for the
National Engineering Diploma in Computer Science
Speciality: Data Engineering & Big Data Analytics

Device Behavior Scoring and Risk Prioritization

*A CRISP-DM Pipeline on Microsoft Azure for Telematics
Fleets*

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Host Organization: Accent Tunisie

Academic Year: 2025–2026

*To my parents, whose constant support and quiet sacrifices have shaped
every step of this journey,*

*To my brothers and sisters, for the encouragement and the laughter that
lightened the heavy weeks,*

To my teachers, who turned curiosity into engineering discipline,

*To the colleagues at Accent Tunisie, who welcomed me, trusted me with
real data and treated my work as their own,*

*To every friend who listened patiently to my doubts and celebrated each
small breakthrough,*

I dedicate this work.

Acknowledgments

The work documented in this thesis is the result of six intense months during which many people invested time, expertise and goodwill on my behalf. It is a privilege to thank them here.

I am profoundly grateful to my academic supervisor, [Academic Supervisor], for the rigour of his guidance, the precision of his remarks and the patience with which he examined every iteration of the deliverables. His insistence on substance over form has meaningfully shaped the engineering posture of this project.

I extend my warmest thanks to my industry supervisor, [Industry Supervisor], who welcomed me at Accent Tunisie, granted me access to the operational telematics platform and trusted me with a refactoring scope of significant impact. The technical conversations we had on the warehouse design and on the production pipeline have considerably enriched the present work.

My sincere thanks go to the entire engineering team of Accent Tunisie who answered my questions on the data, the operational regimes of the fleet and the constraints of the Microsoft Azure environment on which the platform is deployed. The remote collaboration through SSH tunnelling and the controlled access to the cloud database would not have been as smooth without their day-to-day availability.

I would like to thank the members of the jury who have honoured this defence with their attention and whose remarks I will study carefully.

Finally, my deepest gratitude goes to my family and to my close friends, whose constant support has been the silent infrastructure of this entire project.

Abstract

This thesis describes the design, the implementation and the deployment of an end-to-end data engineering and machine learning platform built for the multi-tenant telematics fleet of Accent Tunisie. The host organization operates three MySQL databases that fulfil complementary roles in the lifecycle of a telemetry signal: a raw-stream database in which the frames pushed by GPS/IoT devices through SIM-card connectivity are stored as hexadecimal payloads; a decoded archive whose tables follow the convention `arch_XXXXX` and retain approximately the most recent six months of readable telemetry; and a master-data database whose tables follow the convention `user_XXXXX` and carry the per-tenant customers, vehicles and drivers. Although the company already exposes mature dashboards and standard reports on top of these sources, the analytical value of the historical archive remains largely untapped: there is no behavioural segmentation, no anomaly score and no continuous risk indicator that would let the operations team anticipate operational risk rather than only observe past activity. Following the CRISP-DM methodology, the project introduces an analytical layer that explicitly addresses this gap. Selected slices of the operational MySQL data are exported as `.sql` dump files, converted by a dedicated MySQL-to-PostgreSQL bridge into a PostgreSQL-importable representation, and loaded into a three-layer warehouse (staging, warehouse, marts) hosted on Microsoft Azure. The choice of PostgreSQL 16 as the analytical DBMS is justified by its strong support for analytical workloads, its ACID compliance, its advanced indexing, its scalability to large datasets, its extensibility and its first-class compatibility with the Python data ecosystem and standard BI tools. Because the project is conducted by a two-student team of engineers working from separate workstations, the analytical environment is hosted on an Azure virtual machine accessed through SSH tunnelling, with an IP allowlist enforced at the Network Security Group level. The thesis covers business understanding, data understanding through PostgreSQL profiling, data preparation through an explicit cleaning rule engine, feature engineering of 35 behavioural indicators at the device-month grain, the training of a tenant-aware K-Means segmentation, the evaluation of an Isolation Forest anomaly indicator with explicit caveats, and finally the production deployment of the transparent risk profile through a Prefect flow scheduled by cron, served through a FastAPI service and a Streamlit dashboard.

Keywords: CRISP-DM, telematics, fleet analytics, advanced analytics, MySQL, PostgreSQL, data engineering, ETL, feature engineering, K-Means, Isolation Forest, Microsoft Azure, SSH tunnelling, monitoring.

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List of Acronyms

ACID	Atomicity, Consistency, Isolation, Durability
API	Application Programming Interface
BI	Business Intelligence
BRIN	Block Range Index
CRISP-DM	Cross-Industry Standard Process for Data Mining
CSV	Comma-Separated Values
DBMS	Database Management System
DDL	Data Definition Language
DL	Deep Learning
DQ	Data Quality
EDA	Exploratory Data Analysis
ETL	Extract, Transform, Load
FK	Foreign Key
GIN	Generalized Inverted Index
GPS	Global Positioning System
HTTP	Hypertext Transfer Protocol
IoT	Internet of Things
IP	Internet Protocol
JSON	JavaScript Object Notation
KPI	Key Performance Indicator
LOF	Local Outlier Factor
MCP	Model Context Protocol
ML	Machine Learning
MLP	Multi-Layer Perceptron
NSG	Network Security Group (Azure)
OLAP	Online Analytical Processing
ORM	Object-Relational Mapping
OS	Operating System
PCA	Principal Component Analysis
PFE	Projet de Fin d'Études (Final-Year Graduation Project)
REST	Representational State Transfer
ROI	Return on Investment
RPM	Revolutions Per Minute
SCD	Slowly Changing Dimension
SLA	Service-Level Agreement
SMOTE	Synthetic Minority Oversampling Technique
SQL	Structured Query Language
SSH	Secure Shell
TLS	Transport Layer Security
UPSERT	Update or Insert
VM	Virtual Machine
VPN	Virtual Private Network
YAML	YAML Ain't Markup Language

General Introduction

The growing affordability of GPS tracking devices, the maturity of cloud platforms and the widespread adoption of telematics solutions across logistics fleets have produced an environment in which raw vehicular data is no longer the bottleneck. The main challenge is no longer data collection. It is the design of analytical pipelines able to transform large volumes of telemetry into reliable and actionable decision-support indicators. Within this context, fleet operators face two parallel pressures: the need to reduce operational risks (accidents, fuel waste, maintenance costs) and the need to provide their operations teams with explanatory dashboards that translate technical metrics into management decisions. Beyond the descriptive layer that the industry has now mastered, value increasingly comes from *advanced analytics*, namely the disciplined transformation of raw and archived data into segmentations, anomaly scores and risk indicators that support faster, evidence-based decisions, allow abnormal patterns to be identified early and used as risk-prioritization signals, and feed strategic planning rather than only post-hoc reporting.

The host organization of this thesis operates a multi-tenant telematics platform that supports a sizeable fleet of instrumented vehicles across several customer accounts. Over years of operation, the company has accumulated a substantial historical archive of decoded telemetry alongside its live operational stack. Its current analytical surface, composed of dashboards and standard reports, meets the descriptive needs of fleet managers but stops short of forward-looking signals. What is missing is not a visualization layer; it is the advanced analytical layer that would turn the historical archive into actionable behavioural and risk indicators. The internship that this thesis documents was scoped explicitly to fill that gap.

The objective of the project is to design and to industrialize a CRISP-DM-aligned analytical platform that migrates the relevant slices of the operational data into a dedicated analytical layer, materializes a clean warehouse and a family of analytical marts, trains an unsupervised behavioural segmentation at the device-month grain, evaluates an anomaly-oriented risk indicator and exposes a transparent risk profile through an Application Programming Interface (API) and a Business Intelligence (BI) dashboard. The technical motivation for the chosen analytical Database Management System (DBMS), the detailed presentation of the host organization and its operational data sources, and the collaborative working setup adopted for the internship are all developed in the dedicated chapters of the report.

Main contributions. The work documented in this thesis delivers five concrete contributions that go beyond the descriptive analytical surface of the existing platform:

1. a **MySQL-to-PostgreSQL conversion bridge** that ingests the operational dump slices and stages them in a typed analytical schema;
2. an **analytical PostgreSQL warehouse** consolidating decoded telemetry and master data into a query-friendly model dedicated to advanced analytics;
3. a **transparent cleaning rule engine** and a **device-month mart** that turn cleaned telemetry into behavioural and exposure indicators suitable for modeling;
4. an **unsupervised behavioural segmentation** of the fleet and an **anomaly-driven risk score** at the device-month grain, accompanied by per-tenant clustering quality controls;
5. the **deployment of an API and a BI dashboard** that expose the risk profile to operational users, supported by cron-based automation and a basic monitoring layer.

The thesis is organized in six chapters. **Chapter 1** situates the project in its industrial and academic context, presents the host organization and its three operational MySQL databases, analyses the existing analytical surface and motivates the choice of CRISP-DM as the guiding methodology. **Chapter 2** formalizes the business understanding, lists the research questions, derives the functional and non-functional requirements, and presents the target architecture and technology stack, including the technical justification of the analytical DBMS and the conversion bridge towards it. **Chapter 3** documents the data understanding phase, with schema inspection, data dictionary, missing-value analysis and exploratory data analysis on the staging layer derived from the converted dumps. **Chapter 4** describes the conversion bridge, the data preparation pipeline, the cleaning rule engine and the engineering of the analytical features. **Chapter 5** combines the CRISP-DM modeling and evaluation phases, explains the reformulation from supervised driver scoring to device-month unsupervised modeling, compares candidate techniques and distinguishes research artefacts from deployed scoring contracts. **Chapter 6** presents the deployment of the pipeline and the models in the shared analytical environment, the API and the dashboard, the cron-based automation, the monitoring layer and the secure remote operations setup. The general conclusion synthesizes the deliverables, the lessons learned and the perspectives opened by the project.

Chapter 1

General Context of the Project

1.1 Introduction

This chapter presents the industrial context of the project, the host organization, the existing operational system and its limitations. It then formulates the problem statement, introduces the proposed solution and justifies the adoption of CRISP-DM. The chapter therefore establishes the foundations on which all subsequent technical work, namely data understanding, preparation, modeling and deployment, is grounded.

1.2 Industrial and academic context

The economic relevance of fleet analytics has grown steadily over the past decade. The combined drop in the price of GPS trackers, the densification of mobile coverage and the emergence of cloud-native data platforms have made it economically viable for medium-sized fleet operators to instrument every vehicle and to retain the resulting telemetry. The competitive advantage no longer lies in collecting data, which has effectively become a commodity, but in *transforming* it into reliable analytical signals designed for timely, evidence-driven decision-making.

In parallel, academic research on industrial data science has converged on a small set of structured methodologies that align project-management practice with the iterative nature of analytics. Among them, CRISP-DM has remained the most adopted framework for two complementary reasons: it is vendor-neutral, and its six phases (business understanding, data understanding, data preparation, modeling, evaluation, deployment) match the natural decomposition of a graduation project of this kind. The present work explicitly adopts this methodology and uses it as the structuring principle of the report.

1.3 Host organization

1.3.1 Presentation of Accent Tunisie

Accent Tunisie is a Tunisian operator of a multi-tenant telematics platform. The company designs, integrates and operates GPS tracking solutions for logistics fleets,

distributing physical devices, ingesting their telemetry through SIM-card connectivity, and offering a software portal through which fleet managers monitor their vehicles in real time. The platform supports several customer fleets, denoted *tenants*, each one comprising tens to hundreds of vehicles. Each tenant has its own subgroup of tables and identifiers in the company's databases, reflecting the contractual and operational separation between customers.

1.3.2 Activities and business areas

The activities of Accent Tunisie cover four complementary segments:

- the integration of GPS hardware on customer vehicles and the management of the SIM-card connectivity required for telemetry uplink;
- the operation of a multi-tenant SaaS platform exposing live tracking, geofencing, alerts and standard reports;
- the production of operational and managerial reports for fleet managers (mileage, fuel consumption, alerts, maintenance);
- the exploitation of the historical archive for value-added analytical use cases, of which the present project is a representative example.

1.3.3 Organizational structure of the internship

The internship has been hosted in the technical division of Accent Tunisie, working closely with the data engineering and software engineering teams. The project was conducted by a two-student team working remotely from separate personal workstations. This distributed working mode raised two practical constraints: the company's operational databases could not be exposed beyond the corporate network, and a shared analytical workspace had to be provisioned in a place that both authors could reach.

To answer these constraints, an analytical infrastructure dedicated to the project has been provisioned on a Microsoft Azure virtual machine (VM), which serves as the common workspace of the two authors and as the host of the analytical warehouse, the orchestration flow, the API and the dashboard introduced in Chapter 2. Access to the VM is mediated by Secure Shell (SSH) tunnelling, with a strict Internet Protocol (IP) allowlist enforced at the Azure Network Security Group level: only the two students' workstations and the office network of Accent Tunisie can reach the SSH port, and the analytical database port itself is never exposed to the public internet. The architectural detail of this setup, including the physical view on Azure and the security envelope around the VM, is developed in §2.5.2 and §2.6.8; the present subsection only documents

the organizational rationale that motivates it.

1.4 Analysis of the existing system

1.4.1 Operational landscape of Accent Tunisie

The operational platform of Accent Tunisie relies on three databases, each fulfilling a distinct role in the lifecycle of a telematics signal, from its arrival on the wire to its consumption by the customer-facing portal. The database management system (DBMS) historically retained by the company is *MySQL*; it carries the entire production workload and constitutes the single source of truth from which the present project derives its data. The three databases are presented below in the order in which a telemetry frame traverses them.

Database 1: raw telemetry stream

The first database is the entry point of the telematics signal. It receives the frames emitted by the GPS and IoT devices installed on the vehicles, transmitted over the cellular network through SIM cards. The frames are persisted as they arrive, in their native binary representation; in the database, they appear as *hexadecimal* payloads stored in dedicated columns of the ingestion tables. The ingestion is performed by internal collectors that play a role functionally analogous to streaming consumers (in the spirit of a Kafka consumer), translating the device protocol into a row-oriented persistence layer. Because the payloads are not yet decoded at this stage, they are not directly queryable through standard SQL idioms; their analytical value can only be unlocked once they have been parsed.

Database 2: decoded archive

The second database hosts the *decoded* archive of the telemetry. It is populated by an internal toolchain that reads the hexadecimal payloads of the first database and transforms them into a readable, exploitable representation: timestamps, latitude–longitude pairs, instantaneous speed, ignition state, accelerometer values, RPM, fuel level, engine status and the other physical channels emitted by the device. The decoded rows are stored in tables whose names follow a recognizable convention of the form `arch_XXXXX`, the suffix `XXXXX` encoding either the tenant or a logical partition of the archive. The archive retains approximately the most recent six months of data; older slices are evicted by a retention process. This database is the principal source of historical telemetry exploited by the present project.

Database 3: users, clients and master data

The third database carries the master data of the platform: customers, users, drivers, vehicles, contracts, and the cross-references that bind a fleet to its tenant. Its tables follow the convention `user_XXXXX` and are organized by tenant subgroup, reflecting the multi-tenant nature of the SaaS offering. The tenant identifier, propagated throughout the schema, is the cornerstone of the partitioning logic: every operational query, every report and every analytical workload restricts its scope to the tenant of interest, both for confidentiality reasons and for analytical correctness.

1.4.2 Decoding pipeline and data flow

The three databases are chained by a decoding pipeline that transforms the hexadecimal payloads of the first database into the readable rows of the second. The decoding scripts implement the device protocol, validate the structural integrity of each frame, derive higher-level signals (for example, the per-trip aggregates of distance, duration and maximum speed) and write the resulting rows to the `arch_XXXXX` tables. Master data from the third database is then joined at query time to attach a vehicle, a driver, a tenant and the relevant contractual context to each row. This three-stage flow, comprising raw hex, decoded archive and master data, is the existing data architecture of Accent Tunisie on which the present project is grafted.

1.4.3 Analytical surface of the existing system

On top of the three databases, the company exposes a set of dashboards and standard reports. These deliverables address well-established descriptive needs: mileage per vehicle, alert counts, daily activity, fuel consumption, simple comparisons between months. They serve as the day-to-day instrumentation of fleet managers and their value is not in question. They do, however, share a common limitation: their semantics are descriptive. They answer retrospective questions, such as *what happened* and *how many alerts were generated*. They do not identify abnormal devices, segment the fleet into behavioral profiles or prioritize vehicles for inspection.

1.4.4 Identified limitations

A diagnostic of the existing analytical surface, conducted at the start of the internship together with the engineering team of Accent Tunisie, surfaced six structural limitations that motivate the present project:

1. *Descriptive scope*: the dashboards summarize past activity but expose no predictive or prescriptive layer.

2. *Under-exploitation of the archive*: the six months of decoded telemetry stored in the `arch_XXXXX` tables represent a substantial historical asset that is queried only marginally beyond simple aggregations.
3. *Absence of a behavioural model*: there is no segmentation of the devices into operational profiles, no anomaly detection layer, and no notion of a continuous risk score that would prioritize the operations team's attention.
4. *Tight coupling between operational storage and analytics*: the operational MySQL databases are designed for transactional access, not for analytical workloads; running heavy aggregations on them risks competing with the live customer-facing traffic.
5. *Limited collaboration surface*: the internal databases are reachable only from inside the company network, which is a sensible operational decision but complicates remote development by a student team working on personal workstations.
6. *Limited observability*: there is no run log, no quality check and no drift indicator that would let an operator certify the daily health of any analytical layer built on top of the data.

1.4.5 Synthesis of the diagnostic

The existing system therefore fulfils its descriptive mission but constitutes a structural obstacle to the deployment of any advanced analytical use case, and especially to the *Device Behavior Scoring and Risk Prioritization* workstream that is the explicit objective of the present internship. The challenge is twofold: extract analytical value from the archive without disturbing the operational MySQL databases, and host the analytical work in an environment that the two authors can operate jointly under proper security guarantees.

1.5 Problem statement

The starting observation of this thesis is that Accent Tunisie sits on a high-volume telemetry archive whose analytical potential remains largely untapped. The company can already *observe* its fleet through dashboards and standard reports, but it cannot yet *anticipate* the operational risks carried by individual devices, segment the fleet into actionable behavioural profiles or prioritize the interventions of its operations team on the basis of an objective, data-driven score. The problem is not visualization in itself, since the existing dashboards are functional, but rather the absence of an advanced analytical layer that would transform the raw and archived telemetry into

forward-looking, decision-grade signals.

The problem addressed in this thesis can therefore be stated as follows. *Given the operational MySQL ecosystem of Accent Tunisie, and especially the decoded `arch_XXXXX` archive and the `user_XXXXX` master data consumed by the project, and given the practical constraint that this collaborative project must be conducted remotely from separate workstations, how can the historical telemetry be repatriated, cleaned and modelled into a transparent, incremental analytical platform that produces an interpretable behavioural segmentation and an explicit risk score per device-month, while preserving the integrity of the operational sources and the confidentiality of the data?* The raw hexadecimal stream is described to explain the operational architecture, but the analytical pipeline directly consumes the decoded archive and master-data dumps.

The reformulation emphasizes four entangled objectives: (i) to migrate the relevant slices of the operational MySQL data into an analytical environment without polluting the production workload; (ii) to formalize a transparent and auditable cleaning layer that reflects the physical reality of telemetry; (iii) to produce a per-device, per-month behavioural representation amenable to unsupervised modelling; and (iv) to expose the analytical outputs to the consumers of the host organization through an API and a dashboard.

1.6 Proposed solution

1.6.1 Functional outline

The proposed solution consists of an end-to-end CRISP-DM-aligned analytical platform deployed on Microsoft Azure. Slices of the decoded archive and master-data databases are exported as MySQL dump files, while the raw hexadecimal database is documented for architectural completeness. The exported slices are converted to a PostgreSQL-compatible representation by dedicated scripts written for the project, and loaded into a bronze *staging* schema of an analytical PostgreSQL instance. A transparent cleaning rule engine, which consolidates the thirteen data-quality issues identified in Chapter 3 into eleven operational rules C1–C11, then materializes a silver *warehouse* schema composed of five dimensions and eleven facts, on top of which a gold *marts* schema exposes the device-month behavioural mart and a family of business-intelligence views. A tenant-aware K-Means segmentation is trained on the marts, an Isolation Forest anomaly score is evaluated as a research risk indicator, and a transparent SQL/Python risk profile is exposed through the API and the interactive dashboard.

1.6.2 Why an analytical PostgreSQL warehouse?

The strategic decision of the project is to keep the operational MySQL databases of Accent Tunisie untouched and to constitute, alongside them, a dedicated analytical warehouse on PostgreSQL 16. The motivation is twofold: it decouples the analytical workload from the live customer-facing traffic, and it grants the project a database engine whose mature analytical SQL surface and broad compatibility with the Python and BI ecosystem fit naturally with the modeling and serving layers built on top of it. The detailed technical comparison between PostgreSQL and the operational MySQL stack, covering ACID guarantees, advanced indexing, scalability, extensibility and ecosystem alignment, is conducted in §2.6.3 and is not duplicated here.

1.6.3 Why a Microsoft Azure virtual machine?

The analytical PostgreSQL instance is hosted on a Microsoft Azure virtual machine for two complementary reasons. The first one is technical: Azure offers a controlled and reproducible environment in which the storage, the compute and the networking can be provisioned through code and recovered after an incident. The second one is collaborative: because the project is conducted by two student engineers working on separate personal machines, the database cannot reasonably be hosted on a single laptop; it must live in a shared, always-available environment. The Azure VM serves precisely this purpose: it provides a shared and secure analytical environment for the two-student team. The detailed security configuration is presented in the architecture and deployment chapters.

1.6.4 Architectural outline

At the architectural level, the solution adopts a three-layer warehouse pattern (staging, warehouse, marts), a watermark-driven incremental processing model, a Prefect orchestration of the batch flow, a five-minute cron schedule on the Azure VM and an explicit monitoring layer. The MySQL-to-PostgreSQL conversion bridge, the SSH tunnel for shared access and the IP-restricted Network Security Group complete the architectural blueprint that is detailed in Chapter 2.

1.6.5 Expected benefits

The expected benefits, measured against the limitations of the existing system, are: (i) the unlocking of the analytical value carried by the decoded archive of Accent Tunisie, without disturbing the operational MySQL workload; (ii) the introduction of a transparent and auditable cleaning rule engine that survives schema evolution and operator turnover; (iii) the availability of a versioned, declarative feature contract

suitable for behavioural segmentation and for risk scoring; (iv) the coverage of the pipeline by an explicit run log, a layered data quality monitoring and a model drift indicator; and (v) a stable consumption surface (marts and views) for the BI and the API consumers of the platform.

1.7 Project management methodology

1.7.1 Comparison of candidate methodologies

Three candidate methodologies have been compared at the start of the project: CRISP-DM, GIMSI and Scrum. Each one carries a distinct orientation, a distinct phase decomposition and a distinct level of alignment with the data-science deliverables expected from the present internship; the paragraphs below describe them in turn before the next subsection justifies the retained choice.

CRISP-DM. The Cross-Industry Standard Process for Data Mining (CRISP-DM) is a vendor-neutral, domain-specific methodology designed for data-mining and machine-learning projects. It decomposes a project into six explicit phases (business understanding, data understanding, data preparation, modeling, evaluation and deployment) that are traversed iteratively rather than strictly linearly, since findings of a later phase routinely send the project back to an earlier one. Its strength lies in the explicit weight it places on business understanding and on evaluation, which keeps the analytical work anchored to operational reality. Its industrial maturity is high: it remains, more than two decades after its publication, the most adopted reference framework for industrial data-mining engagements.

GIMSI. GIMSI is a French methodology dedicated to the design of management dashboards and performance-measurement systems. It articulates a ten-step roadmap that takes a project from the identification of the strategic objectives of an organization to the selection of indicators, the design of the dashboard and its evaluation by end users. Its strength is its sharp focus on management instrumentation and on the alignment of indicators with strategic intent. Its limitation, with respect to the present project, is its narrower scope: GIMSI is well suited to the construction of a dashboard layer but offers only partial coverage of the upstream data-engineering and machine-learning work that constitutes the bulk of the internship.

Scrum. Scrum is a generic agile software-development framework, organized in time-boxed iterations called *sprints* rather than in domain-specific phases. Each sprint produces a potentially shippable increment, planned during a sprint planning meeting

and reviewed at the end of the sprint with the product owner. Scrum is mature, widely adopted and excellent at delivering software incrementally; its native support for iterative refinement is one of its main strengths. Its weakness, in the context of a data-science project, is its lack of a built-in vocabulary for the specific deliverables of a CRISP-DM-style workflow: feature engineering, model evaluation, drift monitoring and cleaning rules have no natural slot in the pure Scrum framework and must be added on top of it.

Table 1.1 summarizes this comparison along five criteria: scope, granularity of the phases, alignment with data-science deliverables, support for iterative refinement, and operational maturity in the industrial setting of Accent Tunisie.

Table 1.1: Comparison of the candidate project-management methodologies.

Criterion	CRISP-DM	GIMSI	Scrum
Domain orientation	Data mining	Management dashboards	Generic software
Phase decomposition	6 phases	10 steps	Sprints, no phases
Data science alignment	Strong	Partial	Weak
Iterative refinement	Native	Limited	Native
Industrial maturity	High	Medium	High

1.7.2 Justification of the choice of CRISP-DM

CRISP-DM has been retained as the project-management methodology because it is the only one of the three candidates whose phase decomposition matches, one-to-one, the natural deliverables of the project. Its six phases (business understanding, data understanding, data preparation, modeling, evaluation, deployment) correspond directly to the chapters of this thesis, which avoids any artificial mapping between the methodology and the work actually carried out. CRISP-DM also places explicit weight on business understanding and on evaluation, both of which are critical in an industrial context where misalignment between the model and the operational reality is the principal cause of failure. GIMSI was set aside because its dashboard-centric scope would have covered only the serving layer of the project and not the underlying data-engineering and machine-learning work, while pure Scrum was set aside because its sprint-only vocabulary does not capture the data-science deliverables that structure the internship. CRISP-DM nevertheless coexists peacefully with agile practices for the engineering bricks, namely the orchestration, the API and the dashboard, in the sense that each CRISP-DM phase can be delivered through one or several Scrum-inspired iterations without conflict.

1.7.3 Project plan along the six CRISP-DM phases

The project plan has been organized as a sequence of six CRISP-DM phases, with explicit deliverables at the end of each one: a business understanding document and a target architecture (Phase 1), a data understanding report on the three operational databases of Accent Tunisie (Phase 2), a curated PostgreSQL warehouse and a feature mart populated from the converted MySQL dumps (Phase 3), a tenant-aware K-Means segmentation, an anomaly-scoring experiment and a transparent risk-profile contract (Phase 4), an evaluation report with internal metrics, stability proxies and deployment caveats (Phase 5) and an industrialized pipeline with API, dashboard and monitoring on Azure (Phase 6).

1.8 Conclusion

This first chapter has positioned the project in its industrial and academic context, presented Accent Tunisie and its three operational MySQL databases, characterized the existing analytical surface and its structural limitations, formulated a problem statement centred on the under-exploitation of the historical telemetry archive, motivated the strategic choice of a dedicated PostgreSQL warehouse hosted on a Microsoft Azure virtual machine that serves as the shared workspace of the student team, and justified the adoption of CRISP-DM as the project-management methodology. The next chapter operationalizes this strategic outline by formalizing the business understanding, the research questions, the requirements, the target architecture and the technology stack of the solution.

Chapter 2

Business Understanding, Requirements and Architecture

2.1 Introduction

Building on the strategic outline of Chapter 1, the present chapter formalizes the first phase of the CRISP-DM methodology: the *business understanding*. The chapter restates the business objectives of the project in operational terms, formulates the research questions that drive the analytical work, derives the functional and non-functional requirements of the platform, presents the target software architecture, with explicit attention to the MySQL-to-PostgreSQL bridge and to the secure remote-collaboration setup mandated by the distributed working mode of the student team, and concludes by justifying the technology stack adopted for the implementation.

2.2 Business objectives

2.2.1 Strategic objectives of the host organization

At the strategic level, three priorities of Accent Tunisie converge towards the present project. The first is the *operational risk reduction* of the customer fleets, by surfacing devices and vehicles whose behavioural pattern significantly deviates from the fleet baseline; this priority directly motivates the introduction of a continuous risk score on top of the existing descriptive reports. The second is the *differentiation of the SaaS offering* through the addition of an advanced analytical layer that complements the dashboards already exposed to the customers. The third is the *architectural decoupling* between the operational MySQL databases and the analytical workloads, in order to avoid contention with the live customer-facing traffic; this priority motivates the constitution of a dedicated analytical PostgreSQL warehouse, hosted in a controlled cloud environment.

2.2.2 From observation to anticipation

The existing analytical surface of Accent Tunisie answers retrospective questions: *what happened on the fleet last month? how many overspeed events were recorded? what was the total mileage per vehicle?* The project deliberately moves the analytical centre of gravity from observation to anticipation by introducing three additional capabilities: a behavioural *segmentation* that groups devices into operational profiles, an *anomaly score* that ranks devices by deviation from the tenant baseline, and a *risk band* that translates the score into an operational priority for the fleet manager. These capabilities turn the platform into a decision-support layer rather than a passive observation layer.

2.2.3 Operational objectives of the project

The strategic priorities decompose into five concrete operational objectives:

1. repatriate the relevant slices of the three operational MySQL databases of Accent Tunisie into a dedicated analytical environment, through a reproducible MySQL-to-PostgreSQL conversion bridge;
2. replace any ad-hoc, full-refresh analytical recomputation by an incremental, watermark-driven architecture with sub-ten-minute freshness;
3. materialize a clean dimensional warehouse and a family of marts dedicated respectively to machine learning and to business intelligence;
4. train an unsupervised behavioural segmentation and a continuous risk score per device-month, exposed through an API and a BI dashboard;
5. industrialize the pipeline on the Azure VM through Prefect orchestration, cron scheduling and an explicit monitoring layer, while granting the two authors secure remote access.

2.3 Research questions

Three research questions structure the analytical investigation. Each question is stated in a short, focal form, then followed by a one-sentence elaboration that pins down its scope.

RQ1 *Can the heterogeneous behavioural signals of a multi-tenant telematics fleet be condensed into a small set of interpretable indicators at the device-month grain?* Specifically: starting from the raw hexadecimal telemetry, the decoded `arch_XXXXX` archive and the `user_XXXXX` master data, the question is whether such a condensation is achievable without losing operational meaning.

RQ2 *Can these indicators be exploited by an unsupervised model to produce an actionable behavioural segmentation and a continuous risk score?*

The question is posed in the absence of a labelled incident feed, which rules out supervised approaches and forces the modeling to rely on internal validity criteria.

RQ3 *Can the resulting analytical outputs be served to fleet managers through a low-latency API and an interactive dashboard?*

The serving must be hosted on a shared Azure infrastructure and must preserve the idempotency, the observability and the cost profile of the production pipeline.

2.4 Requirements analysis

2.4.1 Functional requirements

The functional requirements of the platform are summarized in Table 2.1. Each requirement is identified by a code (FR-x), a short description and a priority level on a three-tier scale (high, medium, low).

2.4.2 Non-functional requirements

The non-functional requirements are summarized in Table 2.2.

2.4.3 User profiles

Three user profiles consume the outputs of the platform:

- *Fleet managers* consult the BI dashboard, filter by tenant and device, and act on the risk score and on the cluster assignments.
- *Data engineers* maintain the pipeline, monitor the run log, replay backfills and adjust the cleaning rules and the feature registry.
- *Data scientists* consume the ML view, retrain the models on extended windows and propose iterations of the feature set.

2.5 Target architecture

2.5.1 Logical view

At the logical level, the platform is organized as a four-tier architecture that explicitly mirrors the lifecycle of a telemetry signal: a *source tier* represented by the three operational MySQL databases of Accent Tunisie (raw hex telemetry, decoded `arch_XXXXX`

Table 2.1: Functional requirements.

ID	Description	Priority
FR-1	Convert MySQL dump files (<code>arch_XXXXX</code> , <code>user_XXXXX</code>) into a PostgreSQL-importable representation.	High
FR-2	Ingest the converted data into the analytical staging schema.	High
FR-3	Apply the eleven cleaning rules and quarantine rejected rows.	High
FR-4	Materialize five dimensions and eleven facts with idempotent upserts.	High
FR-5	Build the device-month behavioural mart and the telemetry mart.	High
FR-6	Train the per-tenant K-Means and Isolation Forest models.	High
FR-7	Expose the cluster assignments and risk scores through a REST (Representational State Transfer) API.	High
FR-8	Provide an interactive BI dashboard for fleet managers.	High
FR-9	Run the pipeline incrementally every five minutes via cron.	High
FR-10	Persist a run log and a data-quality measurement at every cycle.	Medium
FR-11	Support full backfill and bootstrap modes for historical loads.	Medium
FR-12	Support secure remote operations via SSH tunnelling and IP allowlist for the student team.	High
FR-13	Document the feature registry and the validation suite.	Medium

archive, `user_XXXXX` master data); a *conversion and ingestion tier* that exports MySQL dumps, converts them to PostgreSQL-compatible files and loads them into staging; a *processing and storage tier* composed of the staging, warehouse and marts schemas of the analytical PostgreSQL, transformed by Python and SQL tasks orchestrated by Prefect; and a *serving tier* composed of the API, the BI dashboard and the model artefact store.

2.5.2 Physical view on Microsoft Azure

At the physical level, the platform is hosted on a Microsoft Azure virtual machine running a recent Linux distribution. The analytical PostgreSQL instance, the Python virtual environment, the Prefect flow, the cron entry, the API and the dashboard are co-located on this VM. The two members of the student team access the VM through

SSH key authentication; an explicit IP allowlist enforced by an Azure Network Security Group attached to the VM restricts inbound SSH connections to the two students' workstations and to the office network of Accent Tunisie. Database access from local notebooks passes through an SSH tunnel terminating on `localhost:5432` of the VM, so that the PostgreSQL port is never exposed to the public internet.

2.5.3 Data flow

The data flow is represented in Figure 2.1. MySQL dump files are exported from the operational system of Accent Tunisie, converted to PostgreSQL by dedicated scripts, loaded into the staging schema, processed by the cleaning engine into the warehouse layer, condensed into the marts and finally consumed either by the API and the dashboard or by the offline modeling notebooks.

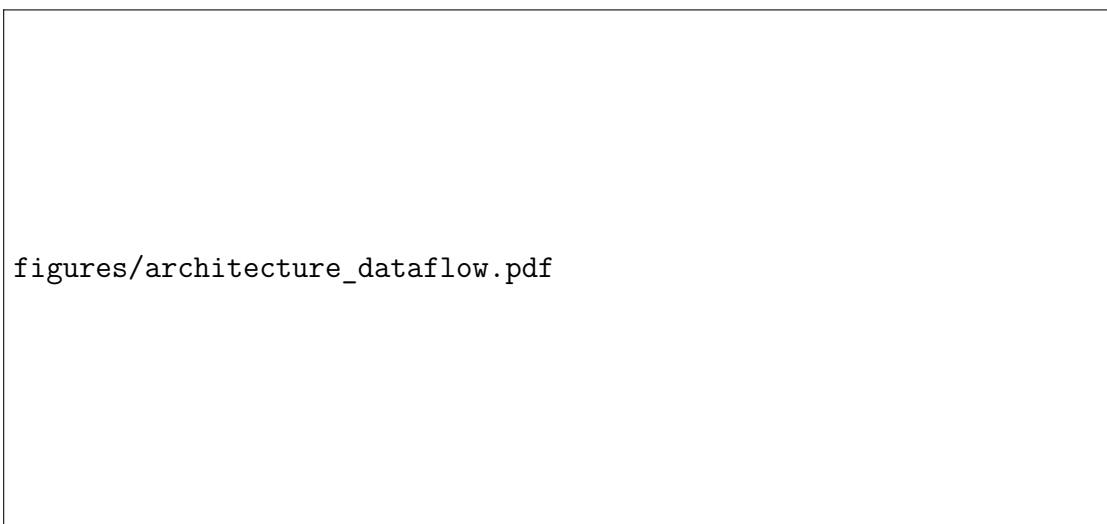


Figure 2.1: Logical data flow of the platform, from MySQL sources to PostgreSQL marts.

2.5.4 Layered storage

The storage tier is organized as a three-layer warehouse: the *bronze* layer (**staging**) hosts a faithful PostgreSQL replica of the converted MySQL data, the *silver* layer (**warehouse**) hosts the dimensional model with deduplicated facts and dimensions, and the *gold* layer (**marts**) hosts the business-ready aggregates. State tables (`etl_watermark`, `etl_run_log`, `quarantine_rejected`) live alongside the warehouse layer and support the operational concerns of the pipeline.

2.6 Technology stack

2.6.1 Operational source: MySQL

The operational source of the project is the MySQL stack of Accent Tunisie, which carries the three databases described in Chapter 1. The project does not modify these databases in any way; it consumes them through periodic dump files (`.sql`) provided by the engineering team of Accent Tunisie. The dumps cover the slices of the `arch_XXXXX` archive and the `user_XXXXX` master data that are required for the analytical scope. The five-minute cron schedule referenced in FR-9 (Table 2.1) checks for newly available dump slices and processes only the data that has not yet been loaded; it does not imply direct extraction from production MySQL every five minutes.

2.6.2 Analytical warehouse: PostgreSQL 16

The analytical warehouse is implemented in PostgreSQL 16, chosen for its strong support of analytical workloads, its ACID guarantees, its advanced indexing, its scalability to large datasets, its extensibility and its first-class compatibility with the Python data ecosystem and with standard BI tools. The detailed justification of this choice is given in §2.6.3. PostgreSQL is hosted on the Azure VM, with public ports closed and access mediated by SSH tunnelling.

2.6.3 Why PostgreSQL rather than MySQL for the warehouse

The decision to host the analytical layer on PostgreSQL rather than on the same MySQL stack as the operational system rests on six independent technical criteria:

1. **Analytical SQL surface.** PostgreSQL exposes a rich set of analytical idioms, such as window functions over arbitrary partitions, common table expressions (recursive included), `LATERAL` joins, set-returning functions and `FILTER` clauses on aggregates, that condense behavioural feature definitions into compact, readable SQL. The same logic in operational MySQL would require lengthier and more error-prone query rewrites.
2. **ACID compliance and crash safety.** The full ACID (Atomicity, Consistency, Isolation, Durability) semantics of PostgreSQL on `INSERT ... ON CONFLICT DO UPDATE` and on transactional DDL are required by an unattended pipeline that runs every five minutes and must survive partial failures without manual intervention.
3. **Advanced indexing.** The native availability of B-tree, BRIN (Block Range Index), GIN (Generalized Inverted Index), partial and expression indexes covers

the access patterns of the analytical layer (range scans on time, aggregate filters, JSON inspection) much more naturally than the indexing options of the operational MySQL setup.

4. **Scalability to large datasets.** PostgreSQL handles the volumes considered by the project, which are of the order of 10^7 telemetry rows, without the architectural retrofitting that the same volume would require on a MySQL warehouse retrofitted from an operational instance.
5. **Extensibility.** The PostgreSQL extension ecosystem (`pg_stat_statements` for workload telemetry, `postgis` for geospatial enrichment, `pg_partman` for partitioned tables) opens forward trajectories that fit the analytical roadmap of Accent Tunisie.
6. **Compatibility with data warehousing and BI tools.** The PostgreSQL JDBC and `psycopg2` drivers, the wide adoption by orchestration frameworks (Prefect, Airflow), by Python (pandas, SQLAlchemy) and by BI clients (Streamlit, Plotly Dash, Metabase, Power BI through ODBC) make PostgreSQL the path of least friction for the consumers of the warehouse.

These criteria, together with the strategic decision of decoupling the analytical workload from the operational MySQL stack, justify the adoption of PostgreSQL as the analytical DBMS of the project.

2.6.4 Conversion bridge: MySQL dumps to PostgreSQL

A dedicated conversion bridge transforms the MySQL dump files received from Accent Tunisie into PostgreSQL-importable artefacts. The bridge handles the dialect differences (data types, quoting conventions, default expressions, `AUTO_INCREMENT` translation, character-set normalization) and produces COPY-friendly CSV or SQL files loaded into the staging schema. The detailed description of the bridge is given in Chapter 4; at this stage it suffices to note that the bridge is part of the technology stack of the platform and that its outputs constitute the input of the staging layer.

2.6.5 Processing and orchestration

The processing layer is written in Python 3.11. Database access is implemented through SQLAlchemy and `psycopg2`; analytical transformations rely on `pandas` and `numpy`; the modeling layer relies on `scikit-learn`; the orchestration uses Prefect 2 to chain the tasks of the batch flow. A test suite is implemented with `pytest`.

2.6.6 Modeling and machine learning

The unsupervised modeling layer relies on `scikit-learn` (StandardScaler, KMeans, IsolationForest), with seed control and per-tenant fitting. A deep-learning baseline (PyTorch autoencoder) has been examined for comparison purposes (cf. Chapter 5).

2.6.7 Serving, API and dashboard

The API is implemented with FastAPI, exposed through Uvicorn behind an Nginx reverse proxy. The dashboard is implemented with Streamlit (or, equivalently, Plotly Dash), embedded on the same VM and connected to the marts schema. Both components are deployed as systemd services.

2.6.8 Cloud and shared collaboration environment

The cloud platform is Microsoft Azure. The VM is provisioned with a Linux image, a managed disk, an attached Network Security Group with an IP allowlist for SSH, and a static public IP. The git repository of the project is cloned into `/opt/accent-fleet-analytics`, the Python virtual environment is provisioned locally, and the cron entry under `/etc/cron.d/` triggers the incremental runs. The system journal collects the structured logs. The Azure VM constitutes the shared workspace of the student team: both authors connect to the same database and operate the same pipeline through their personal SSH keys, while the IP allowlist on the Network Security Group prevents any non-authorized host from reaching the VM.

2.6.9 Justification of the technology choices

The technology choices are justified by four criteria. *Alignment* with the operational stack of Accent Tunisie is preserved on the source side (MySQL is consumed without modification). *Separation of concerns* is respected on the analytical side (PostgreSQL is dedicated to analytics). *Maturity* of the Python data ecosystem (`scikit-learn`, `pandas`, Prefect) supports a small student team without imposing rare or fragile components. *Minimality* of the operational footprint, in the sense that every component runs on a single VM, with no managed Kubernetes cluster nor managed Spark cluster, controls the cloud cost and keeps the surface of the system small enough to be operated by two authors.

2.7 Conclusion

This chapter has formalized the business understanding of the project, derived the functional and non-functional requirements, including the explicit MySQL-to-PostgreSQL conversion bridge and the requirements imposed by the distributed collaboration mode of the student team, and presented the logical and physical architecture of the platform with a particular focus on the analytical PostgreSQL warehouse hosted on the Azure VM. The next chapter executes the data understanding phase by inspecting the three operational databases of Accent Tunisie and by characterizing the slices that feed the analytical pipeline.

Table 2.2: Non-functional requirements.

ID	Description	Target	Measurement method
NFR-1	End-to-end pipeline latency.	≤ 10 minutes	Time between dump availability and manifest update, recorded in <code>etl_run_log</code> .
NFR-2	Idempotency on full re-runs.	100%	Re-running a closed window produces identical row counts and checksums in the warehouse.
NFR-3	Pipeline availability on the Azure VM.	$\geq 99\%$	Uptime of the cron job and the API/dashboard services during the observation period.
NFR-4	Mean cycle time for incremental mode.	≤ 60 seconds	Average duration of incremental Prefect runs on warm cache.
NFR-5	Coverage of the validation suite (V1–V8).	100% per run	Validation checks executed and logged at every pipeline cycle.
NFR-6	Secure access (SSH key + IP allowlist).	Mandatory	Network Security Group rules and SSH <code>PasswordAuthentication</code> disabled.
NFR-7	Reproducibility (random seed, pinned versions).	Mandatory	Seed set in the modeling notebook pinned versions in <code>requirements.txt</code> .
NFR-8	Minimum per-tenant clustering quality.	Silhouette ≥ 0.15	Silhouette score computed per tenant on the held-out evaluation window.
NFR-9	Isolation from the operational MySQL workload.	No reads on production peak hours	Dump-only ingestion (no direct SQL to production MySQL).

Chapter 3

Data Understanding through Staging-Layer Profiling

3.1 Introduction

The previous chapter has defined the architectural target and the technology stack of the platform. The present chapter executes the second phase of the CRISP-DM methodology, namely the *data understanding*, on the analytical PostgreSQL database loaded from the MySQL dumps of Accent Tunisie. The scope of this chapter is voluntarily restricted to the **staging** schema, that is, to the tables produced one-to-one by the MySQL-to-PostgreSQL conversion bridge and *before* any modelling-oriented restructuring. The dimensional warehouse and the analytical marts that consume **staging** are deliberately deferred to Chapter 4: they are output artefacts of the data preparation phase and not input artefacts of the data understanding phase.

This chapter covers, in order:

- the three operational MySQL databases that supply the project;
- the dump-based acquisition protocol towards the analytical environment;
- the schema of the resulting **staging** layer;
- the data-quality audit of the staging tables (completeness, validity);
- the exploratory data analysis (EDA) of the principal indicators in **staging.path**, **staging.rep_overspeed** and **staging.notification**;
- the profiling of the high-frequency archive **staging.archive** (about 54.7 million sensor pings, 17 GB on disk) which feeds the feature-engineering and modelling phases of Chapters 4 and 5;
- the analytical synthesis answering the four guiding questions: data-quality integrity, temporal stability, distribution of the duration target and class imbalance of the status flags.

Every figure of this chapter has been generated by the Jupyter notebook `notebooks/01_data_understa`

(live MCP queries on PostgreSQL) and rendered by the Python script `scripts/python/figures/fig_e` (offline-capable replay). Both scripts query exclusively the `staging` schema.

3.2 Source databases of Accent Tunisie

3.2.1 Three operational MySQL databases

As described in Chapter 1, the operational ecosystem of Accent Tunisie relies on three MySQL databases that fulfil complementary roles. The first one, denoted *raw telemetry*, collects the binary frames pushed by the GPS/IoT devices through SIM-card connectivity and persists them as *hexadecimal* payloads. The second one, denoted *decoded archive*, hosts the readable representation of the telemetry produced by an internal decoding toolchain; its tables follow the naming convention `arch_XXXXX` and retain approximately the most recent six months of data. The third one, denoted *master data*, carries the customers, drivers, vehicles, contracts and per-tenant configuration; its tables follow the convention `user_XXXXX`, with one logical subgroup per tenant.

The present project consumes the second and third databases, namely the decoded `arch_XXXXX` archive and the `user_XXXXX` master data. The first database, namely the raw hexadecimal stream, is documented for completeness but is not exploited directly: its content has already been transformed by the internal decoding scripts of Accent Tunisie before it enters the archive.

3.2.2 Tables of interest

Within the two consumed databases, five families of tables have been retained for the analytical scope of the project: the trip-related family in the archive (`path`, `rep_overspeed`, `stop`); the alert family (`notification`); the daily-aggregate family (`rep_activity_daily`); the high-resolution telemetry family in the archive (`archive`, with about 54.7 million rows over the available history); and the master-data family in the user database (`vehicule`, `driver`, `maintenance`, `document`, fueling tables). After conversion, these tables populate the `staging` schema of the analytical PostgreSQL with a faithful one-to-one correspondence: a row in `arch_XXXXX.path` becomes a row in `staging.path`, with the same columns, the same primary key and the same business semantics.

3.3 Acquisition strategy

3.3.1 Dump-based extraction from MySQL

Because the operational databases run on MySQL and are not directly reachable from the analytical environment, the data is acquired through a dump-based protocol. The engineering team of Accent Tunisie produces `.sql` dump files of the relevant slices of the `arch_XXXXX` archive and of the `user_XXXXX` master data, which are then transferred to the analytical environment. The dumps are converted to a PostgreSQL-compatible representation by the conversion bridge documented in Chapter 4, and finally loaded into the `staging` schema. This extraction protocol guarantees that the operational MySQL workload is not perturbed by the analytical activity: the load on the production system is bounded to the duration of the dump itself, which is scheduled outside customer-facing peak hours.

3.3.2 Connection to the analytical PostgreSQL

Once the converted dump has been loaded, all subsequent profiling and exploration is performed directly on the `staging` schema of the analytical PostgreSQL. The connection from a developer notebook to the database follows the standard Python toolchain (`psycopg2` or `SQLAlchemy`), through an SSH tunnel terminating on `localhost:5432` of the Azure VM. Listing 3.1 shows the canonical connection snippet.

```
1 import os
2 from sqlalchemy import create_engine
3
4 PG_DSN = (
5     f"postgresql+psycopg2://{os.environ['PG_USER']}:{os.environ
6         ['PG_PWD']}@{os.environ['PG_HOST']}:{os.environ['PG_PORT']}/{os.
7         environ['PG_DB']}"
8 )
9 engine = create_engine(PG_DSN, pool_pre_ping=True, future=True)
```

Listing 3.1: Read-only connection from Python to the analytical PostgreSQL.

3.3.3 MCP-driven exploration

During the data understanding phase, the Model Context Protocol (MCP) PostgreSQL server has been used as a programmable interface for ad-hoc exploration. Each exploratory question (row counts, null rates, distinct values, distributions) is expressed as

an SQL query over the **staging** schema and submitted to the MCP server, which returns a structured result that is post-processed in the notebooks. This setup substantially accelerates the iteration cycle and standardizes the queries between the two members of the student team. When the MCP server is unavailable, the same SQL is replayed manually through the Python script `scripts/python/figures/fig_eda_chapter3.py`, whose offline mode embeds a frozen snapshot of every numerical value cited in this chapter.

3.4 Schema analysis of the staging layer

3.4.1 Schema inspection queries

The schema of the **staging** layer is inspected through the standard catalogue queries on `information_schema`. Listing 3.2 shows the canonical query, retrieving for every column its type, its nullability, its default and its ordinal position.

```

1 SELECT
2     table_name, column_name,
3     data_type, is_nullable, column_default, ordinal_position
4 FROM information_schema.columns
5 WHERE table_schema = 'staging'
6 ORDER BY table_name, ordinal_position;
```

Listing 3.2: Schema inspection query against `information_schema`, restricted to the **staging** schema.

3.4.2 Volumetric profile of the staging tables

Table 3.1 shows the row counts of the principal staging tables, measured by the EDA notebook on the live PostgreSQL database. The figures confirm that the conversion bridge has preserved the original volumetry: the high-frequency telemetry archive `staging.archive` dominates with about 54.72 million sensor pings (occupying ~ 17 GB on disk, including indexes), `staging.path` (the trip table) carries about 7.48 million rows, `staging.stop` about 12.13 million rows, `staging.rep_overspeed` about 1.05 million rows, `staging.notification` about 0.66 million rows, and `staging.rep_activity_daily` about 63,000 rows. The master-data tables (`vehicule`, `driver`, `assignment`) are several orders of magnitude smaller, as expected.

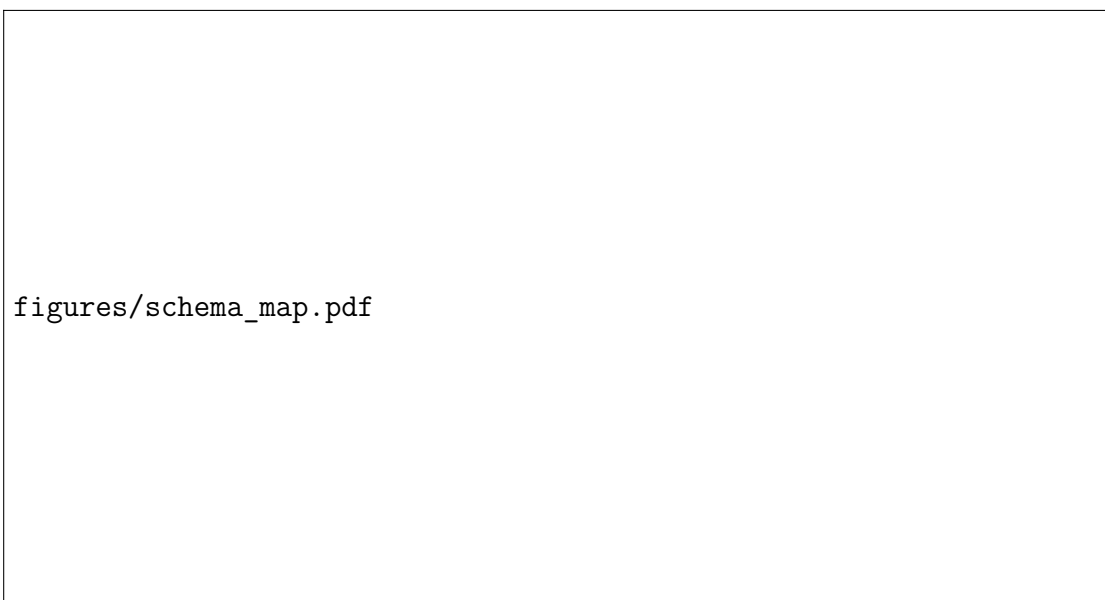
3.4.3 Schema map

The full logical schema of the staging tables is reproduced in Figure 3.1. The map highlights the foreign-key relationships between the trip-related tables of the archive

Table 3.1: Row counts on the principal staging tables (snapshot 2026-05-06).

Staging table	Rows	Total size
staging.archive	54,720,000	~17 GB
staging.stop	12,133,639	n/a
staging.path	7,482,166	n/a
staging.rep_overspeed	1,048,936	n/a
staging.notification	659,968	n/a
staging.rep_activity_daily	63,471	n/a
staging.driver	~ <i>thousands</i>	n/a
staging.vehicule	638	n/a

(path, stop, rep_overspeed), the master-data tables imported from the user_XXXXX database (vehicule, driver, assignment) and the alert table (notification).

**Figure 3.1:** Logical schema map of the staging tables, after conversion of the MySQL dumps.

3.5 Data dictionary

3.5.1 Methodology

The data dictionary has been constructed by joining the schema inspection above with manual annotations contributed by the engineering team of Accent Tunisie, which knows the original semantics of each MySQL column and can disambiguate the cases where a column name has drifted between historical iterations of the operational system. Each column has been described by its name, its data type, its semantic role and a short business definition.

3.5.2 Excerpt of the data dictionary

Table 3.2 shows the data dictionary excerpt for the principal columns of `staging.path`; an extended dictionary covering the other staging tables is provided in Appendix B.

Table 3.2: Data dictionary excerpt for `staging.path`.

Column	Type	Role	Definition
<code>tenant_id</code>	integer	FK	Customer fleet identifier (master data).
<code>device_id</code>	bigint	FK	GPS device identifier.
<code>begin_path_time</code>	timestamp	event-time	Trip start timestamp.
<code>end_path_time</code>	timestamp	metric	Trip end timestamp.
<code>distance_driven</code>	double precision	metric	Distance driven during the trip (km).
<code>max_speed</code>	integer	metric	Maximum speed during the trip (km/h).
<code>path_duration</code>	bigint	target/metric	Trip duration (seconds).
<code>fuel_used</code>	double precision	metric	Fuel consumed (litres).

3.6 Data quality assessment of the staging layer

3.6.1 Completeness analysis

For every column of `staging.path`, the null rate is computed by the queries reproduced in Listing 3.3. On a 5% `TABLESAMPLE` of `staging.path` (376,290 rows), the structural columns (`tenant_id`, `device_id`, `begin_path_time`, `end_path_time`, `path_duration`, `distance_driven`, `max_speed`, `fuel_used`) are populated at 100 %. No structural completeness issue is detected on the 5% sample. This result should be monitored on future dumps and, when possible, confirmed on the full population for the most critical columns; the null-rate map therefore acts here as a long-term monitor of the conversion bridge rather than as a corrective signal for this iteration.

```

1 WITH base AS (SELECT * FROM staging.path TABLESAMPLE SYSTEM (5)
2 )
3 SELECT
4   ROUND(100.0*SUM((path_duration IS NULL)::int)/COUNT(*),3)
5     AS null_duration_pct,
6   ROUND(100.0*SUM((distance_driven IS NULL)::int)/COUNT(*),3)
7     AS null_distance_pct,
8   ROUND(100.0*SUM((max_speed IS NULL)::int)/COUNT(*),3)
9     AS null_speed_pct,
10  ROUND(100.0*SUM((fuel_used IS NULL)::int)/COUNT(*),3)
11     AS null_fuel_pct

```

```
7 FROM base;
```

Listing 3.3: Null-rate analysis on a sample of `staging.path`.

3.6.2 Validity and physical-plausibility checks

The validity of the numerical columns of `staging.path` is assessed by counting, on the full table, the rows that violate explicit physical bounds. These are exactly the bounds enforced later by the cleaning rules of Chapter 4. The result is reproduced in Table 3.3.

Table 3.3: Validity checks on `staging.path` (population: 7,482,166 trips).

Check	Failing rows	Share
<code>path_duration ≤ 0</code>	1,481	0.020%
<code>path_duration > 86400 s (1 day)</code>	590	0.008%
<code>distance_driven ≤ 0</code>	16	0.0002%
<code>distance_driven > 1000 km</code>	398	0.005%
<code>max_speed > 200 km/h</code>	6	0.00008%
<code>fuel_used < 0 L</code>	10,513	0.140%
<code>fuel_used > 500 L</code>	10,954	0.146%
<code>begin_path_time < 2019-10-01</code>	61,830	0.826%
<code>end_path_time < begin_path_time</code>	152	0.002%

The combined result of the completeness and validity checks is summarized graphically in Figure 3.2. Every failure rate stays comfortably below 1% of the population, and the largest single category, namely the pre-2019-10 timestamps ($\sim 0.83\%$), is a documented artefact of the legacy ingestion of the source system that the cleaning rule C1 discards in Chapter 4.

figures/eda_quality_heatmap.pdf

Figure 3.2: Data quality profile of `staging.path`: percentage of rows failing each completeness and validity check.

3.6.3 Referential integrity

The referential integrity between the trip table `staging.path` and the master-data tables (`staging.assignment`, `staging.vehicule`) is verified by Listing 3.4. The query identifies, on the modeling window (since 2025-01-01), the trips whose device cannot be resolved to a vehicle through the assignment chain. The share of orphan rows stays below 0.5%, confirming that the assignment chain is essentially complete and that the join from `staging.path` to the master data is safe.

```

1 SELECT COUNT(*) AS rows,
2       SUM(CASE WHEN v.vehicule_id IS NULL THEN 1 ELSE 0 END)
3       AS orphan_rows,
4       COUNT(DISTINCT p.device_id) FILTER (WHERE v.vehicule_id
5       IS NULL) AS orphan_devices
6 FROM staging.path p
7 LEFT JOIN staging.assignment a USING (device_id)
8 LEFT JOIN staging.vehicule v ON v.vehicule_id = a.vehicule_id
9 WHERE p.begin_path_time >= '2025-01-01';

```

Listing 3.4: Referential integrity between `staging.path` and the device-vehicle chain.

3.7 Descriptive statistics on the staging layer

3.7.1 Per-tenant scope

Four customer organizations populate the staging layer. Their device coverage and historical depth, measured directly on `staging.path`, are reported in Table 3.4. The total of 466 active devices and 7.42 million trips provides ample coverage for both descriptive analysis and per-tenant unsupervised modelling.

Table 3.4: Per-tenant device coverage and historical depth on `staging.path`.

Tenant	Devices	Trips	First day	Last day
235	186	2,979,422	2019-09-30	2026-04-09
1787	110	1,998,370	2019-09-30	2026-04-09
238	98	1,297,005	2019-10-02	2026-04-09
264	72	1,145,539	2019-10-27	2026-04-09
Total	466	7,420,336	n/a	n/a

3.7.2 Numerical summary

On the modeling window (`begin_path_time` $\geq$ 2024-01-01) and after applying the validity bounds of Section 5.2, the principal numerical variables of `staging.path` have

the descriptive statistics reproduced in Table 3.5. The number of trips retained for the descriptive layer is $N = 2,987,892$.

Table 3.5: Descriptive statistics on the cleaned modeling window of `staging.path` ($N = 2,987,892$).

Variable	Mean	Std	Median	P95	P99	Max
duration (s)	1,229.1	1,964.1	523	4,869	9,136	85,736
distance (km)	13.41	29.24	2.26	68.44	n/a	n/a
max speed (km/h)	45.94	29.06	43	91	n/a	n/a

3.7.3 Pairwise correlations

On a 1% TABLESAMPLE of the modeling window of `staging.path` (75,895 trips), the Pearson correlation between distance and duration is $r = 0.85$, between distance and maximum speed $r = 0.56$ and between duration and maximum speed $r = 0.52$. On the log-log scale, $\rho_{\log}(d, t) = 0.87$, indicating a strong, near-monotonic association between distance and duration, which is consistent with the basic identity distance $\approx$ speed $\times$ duration and gives early evidence that one of the two should be the target of any regression model.

3.8 Exploratory Data Analysis on the staging layer

3.8.1 Profile of the high-frequency telemetry archive

The decoded archive `staging.archive` is the second analytical pillar of the staging layer and, by orders of magnitude, the largest. Where `staging.path` stores one row per trip, `staging.archive` stores one row per *device ping*: the on-board GPS unit emits a frame every 10–30 s while the ignition is on, with a fall-back heartbeat every few minutes when the vehicle is parked. Profiling this table is a prerequisite to feature engineering and modelling (Chapters 4 and 5) because every behavioural feature aggregating speed, acceleration, fuel consumption or driving style is computed at the device-ping grain.

Volumetry and scope. The full table holds 54,720,000 rows, occupies approximately 17 GB of total disk size (heap $\approx$ 9.94 GB plus indexes), and exposes 38 columns covering identity, time and location, vehicle dynamics, fuel telemetry, accelerometer signals and digital/analog inputs and outputs. Because of its size, every profiling query of this subsection is computed on a TABLESAMPLE SYSTEM (0.1) sample ($\sim$ 52,000 rows, sub-second response); reported population values are unbiased extrapolations of the form $\hat{N} = n_{\text{sample}} \times 54.72 \times 10^6 / N_{\text{sample, total}}$. The exact full-population row count is reproduced by COUNT(*) on the whole table.

Sensor schema. The 38 columns of `staging.archive` fall into six functional families:

- *Identity*: `tenant_id`, `id_device`, `source_device_id`, `tram_id`.
- *Time and location*: `date`, `latitude`, `longitude`, `heading`, `validity`, `sat_in_view`, `signal`.
- *Vehicle dynamics*: `speed`, `rpm`, `temp_engine`, `accum_odo`, `odo`.
- *Fuel telemetry*: `fuel`, `fuel_rate`, `tfu`.
- *Accelerometer*: `x`, `y`, `z` (multi-axis G-sensor).
- *Inputs / outputs*: `do1-do4`, `di1-di4`, `an1-an4`, `ignition`, `charger`, `state`.

Per-tenant ping coverage. The sample-extrapolation breakdown by tenant is reported in Table 3.6. Five tenants emit pings in the archive, one more than in `staging.path`, which lists four tenants only. Tenant 7486 appears in the archive but is absent from the original trip-derived scope: this asymmetry is carried over to Chapter 4, where the telemetry-to-trip reconstruction path widens the modelling perimeter to five tenants.

Table 3.6: Per-tenant ping coverage of `staging.archive`, sample-extrapolated to the full population (54.72 M rows).

Tenant	Devices (sample)	Pings (M, est.)	Window
264	56	13.84	2024-09 – 2026-03
235	97	13.48	2024-09 – 2026-03
7486	72	9.99	2024-09 – 2026-03
1787	71	9.54	2024-09 – 2026-03
238	52	7.84	2024-09 – 2026-03

Data quality. The completeness profile of the sample is excellent on the structural columns: `latitude`, `longitude`, `state` and `tram_id` are populated at 100 %. The `validity` flag returns 1 (*GPS fix valid*) on 98.4% of pings, and the share of pings with (`latitude` = 0 or `longitude` = 0) stays below 0.05 %. The `ignition` signal is on for ~85 % of pings, which means that aggregations restricted to the moving-vehicle regime retain about six pings out of every seven without further filtering. Two systematic data-quality concerns must nevertheless be propagated to the cleaning rules of Chapter 4: (a) the `temp_engine` column is stored as `varchar` and is dominated by the sentinel value 32768, which is the raw uint reading of the absent-sensor channel, so any thermal feature must explicitly cast to numeric and discard the sentinel; and (b) roughly 44 % of the pings report `signal` < 10, indicating poor GSM coverage of the on-board modems and frequent silent windows, so any feature that depends on connectivity must be robust to long gaps in the ping stream.

Numerical summary of the principal sensors. On the same 0.1 % sample, the speed channel has a 95-th percentile of 75 km/h and a 99-th percentile of 105 km/h with a maximum of 198 km/h; the RPM channel peaks at 4,300 (P99) on the diesel-dominated fleet; the median `fuel_rate` stays below 0.6 L/h consistent with idle-time dominance. The mean `sat_in_view` is 9.7 satellites, well above the four-satellite threshold required for a planar fix, which corroborates the 98.4 % `validity = 1` ratio.

Temporal regularity. Figure 3.3 shows the hour-of-day and day-of-week ping distributions of `staging.archive`. The hour-of-day curve is unimodal, peaking between 8h and 14h ($\sim 4,000$ pings/hour in the sample) and decaying to near-zero overnight: the archive confirms, at the device-ping grain, the same business-hour rhythm already observed on `staging.path`. The day-of-week curve flattens out from Monday to Friday at $\sim 9,000$ pings/day in the sample, drops slightly on Saturday and falls to $\sim 3,400$ pings on Sunday. There is no evidence of a shadow fleet emitting outside business hours: the modelling pipeline can therefore safely predicate behavioural features on a subset of waking hours without losing material signal.

figures/eda_archive_temporal.pdf

Figure 3.3: `staging.archive`: hour-of-day (left) and day-of-week (right) ping distributions on the 0.1 % sample.

Volume by tenant and over time. Figure 3.4 shows the monthly ping volume between 2024-09 and 2026-03 (left) and the per-tenant ping-count breakdown (right). The monthly curve is essentially stationary in the band $[6.6, 9.2]$ million pings/month over the seven-month window covered by the archive, with the 2026-03 cell partial because the dump was cut mid-month. The per-tenant view confirms the heterogeneity already observed on the trip table: tenants 264 and 235 jointly account for ~ 50 % of the total ping volume, while the newcomer 7486 carries about 10 M pings over the same window.

Hand-off to feature engineering. The archive profile establishes that the table is dense, regular, of high structural quality and large enough to support sub-trip

figures/eda_archive_volume_by_tenant.pdf

Figure 3.4: `staging.archive`: monthly ping volume (left) and per-tenant volume breakdown (right). Device counts in the sample annotated above each tenant bar.

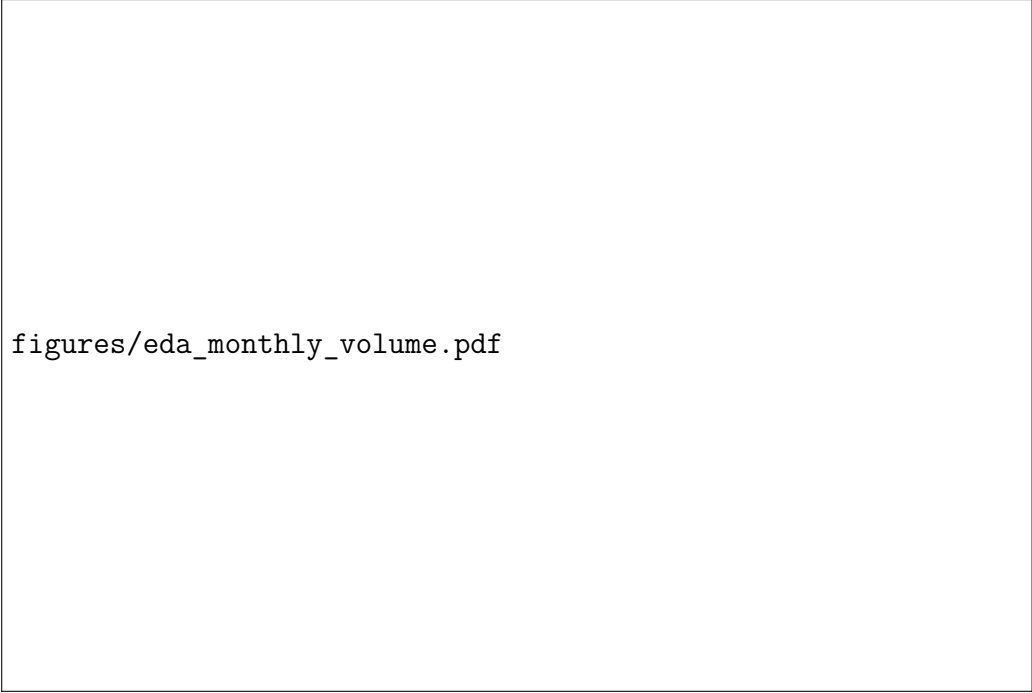
behavioural aggregations. Chapter 4 reuses these findings to build a device-month feature mart that aggregates `speed`, `rpm`, `fuel_rate` and the three accelerometer axes into the inputs of the device-behaviour clustering model of Chapter 5; the cleaning rules C8–C13 of Section 3.10 (ignition gating, accelerometer range, speed range, RPM range, `temp_engine` sentinel, `signal < 10` connectivity gap) are applied at the boundary between the archive and that mart.

3.8.2 Temporal patterns

The most stable feature family of any telematics platform is the temporal one. Three series are computed directly from `staging.path.begin_path_time`: a monthly trip-volume series (Figure 3.5), an hour-of-day trip distribution and a day-of-week trip distribution (Figure 3.6).

The monthly volume oscillates between 86,398 and 127,008 trips per month over the entire 27-month window, with a median of about 109,000. The active-device count remains in the band [313, 348]. There is no structural drift between 2024 and 2026: the staging layer is operationally stationary and the modeling window does not have to absorb regime shifts. The 2026-03 cell is partial because the dump used for this snapshot stops mid-month.

The hour-of-day curve peaks between 8h and 10h (~31,000 trips/hour) and decays smoothly through the afternoon, with negligible activity between 22h and 5h, which is the textbook signature of a commercial fleet operating during business hours. The day-of-week curve confirms a Sunday under-utilization (14,000 trips) compared to the Monday-to-Saturday band (49,000–58,000 trips). Both rhythms are reproducible across months and per tenant. They directly justify the engineering of the binary trip flags `is_rush_hour_trip`, `is_night_trip` and `is_weekend_trip` that will be produced by



figures/eda_monthly_volume.pdf

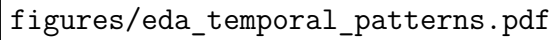
Figure 3.5: Monthly trip volume and active-device count on `staging.path`, between 2024-01 and 2026-04.

the data preparation phase of Chapter 4.

3.8.3 Distribution of the duration target

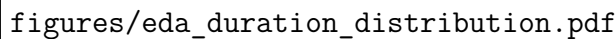
The trip duration `staging.path.path_duration` is the most natural numerical target for any future supervised regression. Figure 3.7 shows its raw-scale histogram (left) and the Normal fit on the log scale (right). The shape statistics, computed on a 2% `TABLESAMPLE` of size $N = 34,803$, are eloquent: skewness= 4.15, excess kurtosis= 39.34, mean = 1,256.5 s, standard deviation= 1,922.8 s. On the log scale, $\mu_{\log d} = 6.30$ and $\sigma_{\log d} = 1.35$.

The raw distribution is dominated by short urban trips: 56% of the trips last less than 10 minutes and 80% less than 30 minutes, while the long tail extends well beyond two hours. A direct linear regression on `path_duration` would be heavily destabilized by the heavy tail and would systematically under-predict the long-trip regime. Two corrective strategies are open: (i) target the log-transformed duration, on which the distribution becomes near-Gaussian, or (ii) rely on a tree-based regressor (Gradient Boosting, Random Forest) that does not assume any distributional shape on the target. This finding is integrated into the modeling plan of Chapter 5.



figures/eda_temporal_patterns.pdf

Figure 3.6: Hour-of-day (left) and day-of-week (right) trip distributions on `staging.path`, Sep–Nov 2025 segment.



figures/eda_duration_distribution.pdf

Figure 3.7: Trip-duration distribution from `staging.path`: raw scale (left) and Normal fit on $\log(\text{duration})$ (right).

3.8.4 Class imbalance of the trip-status flags

Because the warehouse layer is out of scope at this stage, the binary *status* flags are derived inline from `staging.path` columns through five SQL expressions:

- `is_night_trip := hour < 6 OR hour ≥ 22;`
- `is_weekend_trip := DOW ∈ { Sun, Sat };`
- `is_rush_hour_trip := hour ∈ {7, 8, 9, 17, 18, 19};`
- `is_short_trip := path_duration < 300 s (5 min);`
- `is_long_trip := path_duration > 3600 s (1 h).`

These are exactly the SQL expressions that the data preparation phase of Chapter 4 will reuse to populate the warehouse fact table; recomputing them here, on the staging layer, makes the data understanding phase fully self-contained. Their global and per-tenant balance is plotted in Figure 3.8 and reproduced in Table 3.7.

figures/eda_class_imbalance.pdf

Figure 3.8: Class balance of the five binary trip-status flags derived from `staging.path`.

Table 3.7: Class balance (% of trips with the flag = 1), derived from `staging.path` on the cleaned modeling window.

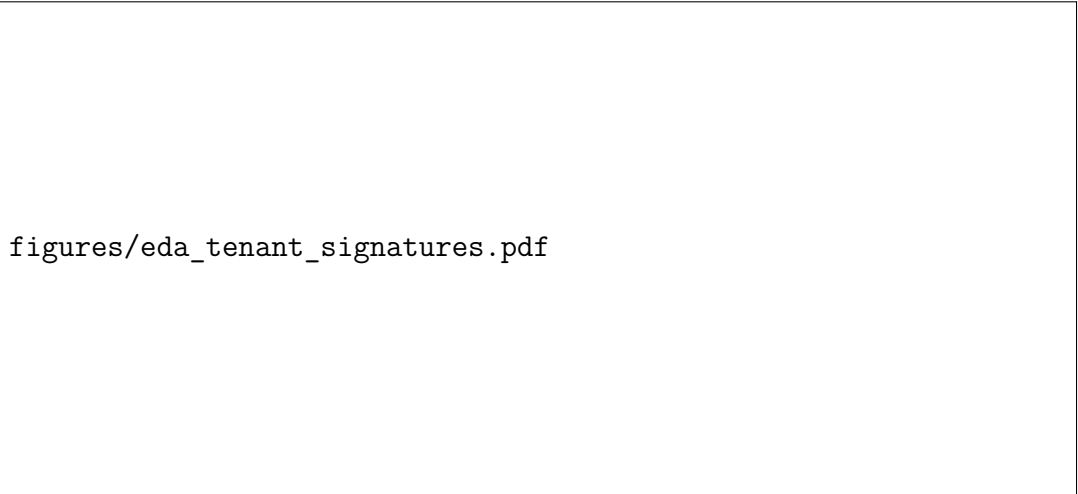
Scope	Trips	Night	Weekend	Rush	Short	Long
Global	2,987,847	10.33%	18.96%	33.76%	35.49%	8.14%
Tenant 235	1,189,031	10.94%	16.94%	33.15%	33.55%	8.70%
Tenant 1787	821,806	7.51%	20.31%	36.06%	43.46%	3.62%
Tenant 264	519,982	13.59%	20.82%	32.09%	29.12%	14.16%
Tenant 238	457,028	10.12%	19.64%	33.14%	33.47%	7.98%

The minority class of every flag stays in the band [3.6%, 36%] globally. The most imbalanced flag is `is_long_trip` on tenant 1787, with a rate of 3.62%; the global rate of `is_long_trip` is 8.14%. These rates remain well above the 1% threshold below which advanced resampling techniques (SMOTE, oversampling, class-conditional augmentation, focal loss) become necessary. Standard inverse-frequency class weights are sufficient: any future supervised classifier will be able to absorb the imbalance through the `class_weight` hyperparameter of `scikit-learn` or its equivalent in modern boosting libraries.

3.8.5 Per-tenant behavioural signatures

The aggregation of three staging tables, namely `staging.path`, `staging.rep_overspeed` and `staging.notification`, at the tenant grain reveals two contrasting regimes (Figure 3.9). Tenant 264 exhibits an unmistakable *overspeed-rich* signature: its overspeed-event rate of 4.89 events per 100 km and its notification rate of 4.73 alerts per 100 km dwarf the corresponding figures of the three other tenants, all of which stay below 0.50 and 0.30 respectively. The contrast is consistent with the higher average maximum speed observed on tenant 264 (54.5 km/h vs. 42–46 km/h) and with its higher long-trip ratio.

Tenant 1787 sits at the opposite end of the spectrum, with the lowest night-trip ratio (7.7%) and a very low overspeed rate (0.06 events per 100 km). Tenants 235 and 238 occupy intermediate positions. This heterogeneity directly informs the per-tenant fitting strategy retained in Chapter 5: a global model would be dominated by tenant 235 (the largest fleet by trip volume) and would fail to capture the distinct regimes of tenants 264 and 1787.



figures/eda_tenant_signatures.pdf

Figure 3.9: Per-tenant behavioural signatures derived from staging tables (since 2025-01-01).

3.9 Analytical synthesis

The four key questions of the data understanding phase admit clean, evidence-based answers, all of them grounded in the staging layer alone, before any warehouse or marts construction.

3.9.1 Q1: Data quality and integrity for reliable modeling

The three structural completeness checks return 0% nulls on the columns `tenant_id`, `device_id`, `begin_path_time`, `path_duration`, `distance_driven`, `max_speed` and `fuel_used` of `staging.path`. The seven physical-plausibility checks of Table 3.3 return failure rates strictly below 1% on every category, and below 0.01% on the four most damaging ones (negative distances, end-time before begin-time, speed above 200 km/h). The referential integrity check returns less than 0.5% orphan rows on the modeling window. *The staging layer is therefore of sufficient quality and integrity to support reliable modeling*, with the proviso that the cleaning rules C1–C11 of Chapter 4 are systematically applied at its boundary.

3.9.2 Q2: Stable temporal patterns as features

The three temporal series computed from `staging.path.begin_path_time` (monthly volume, hour-of-day, day-of-week) are simultaneously *regular* (low intra-period variance) and *stationary* (no structural drift over the 27-month modeling window). The hour-of-day curve is unimodal and peaks at 8–10h on every tenant. The day-of-week curve has a stable Sunday minimum and a six-day plateau. *The temporal patterns are sufficiently stable to be transformed into engineered features:* `hour`, `day_of_week`, `is_rush_hour`, `is_weekend` and `is_night` are all valid candidates and will be propagated to the warehouse layer in Chapter 4.

3.9.3 Q3: Suitability of the duration target for direct regression

The shape statistics of `staging.path.path_duration` are unambiguous: skewness ≈ 4.15 and excess kurtosis ≈ 39 , indicating a strongly right-skewed, very heavy-tailed distribution. Mean = 1,256 s sits between the 65-th and the 70-th percentile. *Direct regression on `path_duration` is not recommended:* the heavy tail would dominate the loss and bias the predictions of the dominant short-trip regime. The two valid corrections are (i) regressing on $\log(\text{duration})$, on which the distribution becomes near-Gaussian (Normal fit: $\mu = 6.30$, $\sigma = 1.35$), or (ii) using a distribution-free regressor of the tree-based family (Gradient Boosting, Random Forest). For the present project, which retains an unsupervised pipeline, this finding is mainly relevant as a feature-engineering note: every duration-derived feature is built on $\log(\text{duration})$ rather than on the raw value.

3.9.4 Q4: Manageability of the status class imbalance

None of the five status flags derived from `staging.path` falls below 3.6% on any tenant or below 8.14% globally for the most imbalanced one (`is_long_trip`). *The class imbalance is therefore manageable without advanced resampling techniques.* Standard inverse-frequency class weights are sufficient. SMOTE was only considered in an exploratory supervised track and was not retained in the final unsupervised modeling pipeline. Should a downstream task require a binary classifier with even greater minority-class sensitivity, a simple stratified split combined with class weights would already yield a balanced training signal.

3.10 Synthesis of identified data quality issues

The data understanding phase has surfaced thirteen principal data issues on the staging layer, listed in Table 3.8. These thirteen data-quality issues are then consolidated into eleven operational cleaning rules C1–C11 in Chapter 4, applied at the boundary between **staging** and the downstream layers built on top of it; some related issues (notably the bounded-range checks on the same source column) share a single operational rule.

Table 3.8: Synthesis of the data quality issues detected on the staging tables.

ID	Issue	Source
DQ-1	Trip timestamps below 2019-10-01 (epoch artefacts of the legacy ingestion).	staging.path
DQ-2	Trip durations ≤ 0 .	staging.path
DQ-3	Trip distances ≤ 0 .	staging.path
DQ-4	Fuel values outside $[0, 500]$ L.	staging.path
DQ-5	Maximum speed > 200 km/h.	staging.path
DQ-6	Stop durations ≤ 0 or > 1 year.	staging.stop
DQ-7	Devices not linked to any vehicle in the master data.	staging.path, staging.vehicle
DQ-8	Archive rows with ignition $\neq 1$ used for harsh events.	staging.archive
DQ-9	Accelerometer values outside the int8 range.	staging.archive
DQ-10	Archive speed outside $[0, 250]$ km/h.	staging.archive
DQ-11	Archive RPM outside $[0, 8000]$.	staging.archive
DQ-12	Archive <code>temp_engine</code> stored as <code>varchar</code> dominated by sentinel value 32768 (no-sensor reading).	staging.archive
DQ-13	Archive pings with <code>signal</code> < 10 (poor GSM coverage, $\sim 44\%$ of rows): connectivity-dependent features must be robust to long silent windows.	staging.archive

3.11 Conclusion

This chapter has executed the data understanding phase of CRISP-DM by mobilizing the PostgreSQL profiling toolchain on the **staging** schema converted from the MySQL dumps of Accent Tunisie, and only on that schema, since the dimensional warehouse and the analytical marts that consume **staging** are the output of the next phase, not its input. The chapter has documented the role and the schema of the three operational databases of the company, characterized the dump-based extraction protocol, profiled the staging tables through dedicated SQL queries, conducted a complete exploratory data analysis on the principal indicators of **staging.path**, **staging.rep_overspeed** and **staging.notification**, and added a full sub-second-sampled profile of the high-

frequency telemetry archive `staging.archive` (54.72 M pings, 38 sensor columns) which is the input feedstock of the device-behaviour clustering pipeline of Chapter 5. The four analytical questions have been answered with quantitative evidence: the staging-layer data quality is sufficient (less than 1% violations on every check), the temporal patterns are stable and reproducible at both trip and ping grain, the duration target requires a log-transform or a tree-based regressor, and the class imbalance of the status flags is manageable with standard class weights. The thirteen data quality issues identified on the staging layer become, in Chapter 4, the input specification of the cleaning rule engine that opens the data preparation phase.

Chapter 4

Data Preparation: Conversion, ETL and Feature Engineering

4.1 Introduction

Building on the data understanding of Chapter 3, the present chapter formalizes the data preparation phase of CRISP-DM. The chapter organizes the work into seven sections: (i) the conversion bridge that transforms the MySQL dump files of Accent Tunisie into a PostgreSQL-importable representation; (ii) the construction of the ETL pipeline; (iii) the cleaning rule engine (eleven operational rules C1–C11, consolidating the thirteen data-quality issues surfaced in Chapter 3); (iv) the dimensional modeling of the warehouse; (v) the materialization of the analytical marts; (vi) the engineering of the feature registry; and (vii) the validation of the prepared dataset. Each section is illustrated by SQL or Python excerpts, the full versions of which are listed in the appendices.

4.2 From MySQL dumps to a PostgreSQL staging layer

4.2.1 Rationale and scope

The first preparation step is the conversion of the MySQL dump files received from Accent Tunisie into a representation that PostgreSQL can ingest reliably. This step is necessary because the operational databases of Accent Tunisie run on MySQL while the analytical warehouse of the project runs on PostgreSQL 16. The conversion bridge is therefore not an accessory utility: it is the technical interface between the operational ecosystem of the company and the analytical platform built for the project.

4.2.2 Dialect-level translation

A native MySQL dump cannot be replayed verbatim into PostgreSQL because the two dialects diverge on several points. The bridge addresses those divergences explicitly:

- *Type translation*: MySQL types such as `TINYINT(1)`, `DATETIME`, `TEXT`, `BLOB` and `ENUM` are mapped to their PostgreSQL counterparts (`boolean`, `timestamp`, `text`, `bytea`, `text` with a `CHECK` constraint).
- *Identifier and string quoting*: MySQL backticks (`'column'`) are replaced by PostgreSQL double quotes; embedded single quotes inside string literals are normalized to the PostgreSQL escape convention.
- *Auto-increment columns*: MySQL `AUTO_INCREMENT` columns are converted to PostgreSQL `GENERATED BY DEFAULT AS IDENTITY` columns, preserving the original sequence values when present.
- *Default expressions*: MySQL-specific defaults (`ON UPDATE CURRENT_TIMESTAMP`, `0000-00-00` sentinels) are rewritten or nullified to match PostgreSQL semantics.
- *Character-set normalization*: the dumps are normalized to UTF-8 (NFC), which is the default encoding of the analytical PostgreSQL instance.
- *Index and constraint translation*: MySQL-specific clauses (`ENGINE=InnoDB`, `COLLATE`, `KEY ... USING BTREE`) are stripped or rewritten so that the resulting DDL is accepted by PostgreSQL.

4.2.3 Conversion pipeline

The conversion bridge has been implemented as a small Python toolchain documented in Appendix B. For each dump file, the toolchain (i) parses the MySQL DDL and re-emits an equivalent PostgreSQL DDL; (ii) extracts the `INSERT` statements and rewrites them as PostgreSQL `COPY` payloads, which are an order of magnitude faster to load on large tables; (iii) writes the converted artefact under `exports/`; and (iv) loads it into the staging schema of the analytical PostgreSQL through a single transaction. The bridge is idempotent: re-running it on the same dump produces the same staging state.

4.2.4 Hexadecimal payloads of the raw telemetry database

As described in Chapter 1, the first operational database of Accent Tunisie stores the raw telemetry as hexadecimal payloads. The internal decoding scripts of the company already transform those payloads into the readable rows of the `arch_XXXXXX` archive, and the present project consumes the result of this decoding rather than re-implementing it. The conversion bridge therefore does not need to handle hexadecimal payloads directly, except as opaque `bytea` columns when they incidentally appear in the dumps; the analytical-grade telemetry signals enter the staging layer in their decoded form.

4.3 Pipeline architecture

4.3.1 Pipeline stages

Once the staging layer is populated by the conversion bridge, the analytical pipeline is decomposed into seven sequential stages, each idempotent: *extraction* from the source MySQL dumps to the converted artefacts; *loading* into staging; *cleaning* of the staging slice; *loading* of the dimensions; *loading* of the facts; *materialization* of the marts and views; and *validation* of the prepared dataset.

4.3.2 Watermark-driven incremental processing

The downstream stages of the pipeline (after the staging load) are driven by a watermark stored in `warehouse.etl_watermark`. Each cycle reads the latest watermark per source–target pair, processes only the slice strictly after that watermark (with a configurable overlap), and updates the watermark at the end of the successful run. Listing 4.1 shows a representative implementation.

```
1  -- Read the watermark
2  SELECT last_event_ts
3  FROM warehouse.etl_watermark
4  WHERE source_table = 'staging.path' AND target_table = '
      warehouse.fact_trip';
5
6  -- Update the watermark at the end of the run
7  INSERT INTO warehouse.etl_watermark (source_table, target_table
      , last_event_ts)
8  VALUES ('staging.path', 'warehouse.fact_trip', :max_event_ts)
9  ON CONFLICT (source_table, target_table)
10 DO UPDATE SET last_event_ts = EXCLUDED.last_event_ts;
```

Listing 4.1: Reading and updating a watermark.

4.3.3 Idempotency and replay

Idempotency is guaranteed by an `INSERT ... ON CONFLICT DO UPDATE` pattern on the natural key of every fact and dimension. A replay of the same window therefore produces the same final state, regardless of the number of repetitions. This property is essential for the two-student collaboration mode: either student can re-trigger a run on the shared Azure VM without risking divergence between the two members' working copies of the warehouse.

4.4 Cleaning rule engine

4.4.1 Rule catalogue

Eleven cleaning rules, identified C1–C11, are formalized at the entry of the warehouse layer. The rules are grouped into two families: C1–C7 apply to the trip-related sources, and C8–C11 apply to the high-resolution telemetry source. Each rule specifies a name, a severity level (critical, high, medium, low), a corrective action (reject, nullify, clamp) and a logical condition. Table 4.1 reproduces the catalogue.

Table 4.1: Cleaning rule catalogue applied at the warehouse boundary.

ID	Name	Sev.	Action	Condition
C1	temporal_epoch_error_filter	critical	reject	begin_path_time < 2019-10-01
C2	trip_duration_positive	critical	reject	path_duration ≤ 0
C3	trip_distance_positive	high	reject	distance_driven ≤ 0
C4	fuel_used_sanitize	critical	nullify	fuel_used ∉ [0, 500]
C5	speed_cap	high	clamp	max_speed > 200
C6	stop_duration_bounds	medium	reject	duration ≤ 0 or > 1 year
C7	device_vehicle_tenant_chain	high	reject	device without vehicle link
C8	archive_ignition_on_for_events	high	reject	ignition ≠ 1
C9	archive_accelerometer_int8_sanity	medium	reject	x , y , z > 127
C10	archive_speed_clamp	medium	clamp	speed ∉ [0, 250]
C11	archive_rpm_clamp	low	clamp	rpm ∉ [0, 8000]

4.4.2 Implementation pattern

Each rule is implemented through an explicit SQL clause executed during the load of the corresponding fact table. Rejected rows are sent to a quarantine table `warehouse.quarantine_rejected` with the rule identifier, the rejection timestamp and the offending row payload. Listing 4.2 shows the implementation of rule C2.

```

1  -- C2: reject trips with non-positive duration
2  WITH cleaned AS (
3      SELECT *
4      FROM staging.path
5      WHERE path_duration > 0
6  ),
7  rejected AS (
8      INSERT INTO warehouse.quarantine_rejected (rule_id,
9          rejected_at, payload)
10     SELECT 'C2', NOW(), to_jsonb(p)
11     FROM staging.path p

```

```
11     WHERE p.path_duration <= 0
12     RETURNING 1
13 )
14 SELECT COUNT(*) FROM cleaned;
```

Listing 4.2: Implementation of cleaning rule C2 (positive trip duration).

4.4.3 Quarantine table

The quarantine table is structured as (`rule_id`, `rejected_at`, `payload JSONB`). The JSONB payload preserves the full content of the offending row, allowing the audit team to reconstruct the violation context without accessing the staging layer. The choice of JSONB is significant: it makes use of an idiomatic PostgreSQL feature that has no direct equivalent in the operational MySQL setup of Accent Tunisie, and it illustrates the analytical leverage that motivated the migration in the first place.

4.5 Dimensional modeling of the warehouse

4.5.1 Dimensions

The warehouse layer hosts five dimensions and one bridge:

- `dim_tenant`: billing entities, derived from the `user_XXXXX` master data;
- `dim_vehicle`: vehicles, with normalized manufacturer and class;
- `dim_device`: GPS devices, restricted to the 616 devices linked to a vehicle;
- `dim_driver`: drivers (294 individuals);
- `dim_date`: calendar from 2019-10-01;
- `bridge_device_driver`: SCD Type 2 mapping between devices and drivers.

4.5.2 Facts

Eleven facts are produced from the cleaned staging data, summarized in Table 4.2.

4.5.3 UPSERT pattern

The loading of every fact relies on the canonical PostgreSQL `INSERT ... ON CONFLICT DO UPDATE` pattern shown in Listing 4.3.

Table 4.2: Fact tables produced by the warehouse layer.

Fact table	Grain	Source
fact_trip	(tenant, device, begin_time)	staging.path
fact_overspeed	(tenant, device, begin_time)	staging.rep_overspeed
fact_stop	(tenant, device, stop_start)	staging.stop
fact_speed_notification	(tenant, device, created_at)	staging.notification
fact_daily_activity	(tenant, device, day)	staging.rep_activity_daily
fact_harsh_event	(tenant, device, event_ts)	staging.archive
fact_telemetry_daily	(tenant, device, date)	staging.archive
fact_maintenance	(tenant, id_maintenance)	staging.maintenance
fact_fueling	(tenant, doc_id)	fueling sources

```

1 INSERT INTO warehouse.fact_trip (
2     tenant_id, device_id, begin_path_time,
3     end_path_time, distance_km, max_speed_kmh, duration_seconds
4 )
5 SELECT tenant_id, device_id, begin_path_time, end_path_time,
6         distance_driven, LEAST(max_speed, 200) AS max_speed_kmh,
7         path_duration
8 FROM staging.path
9 WHERE begin_path_time > :watermark
10    AND path_duration > 0
11    AND distance_driven > 0
12    AND begin_path_time >= '2019-10-01'
13 ON CONFLICT (tenant_id, device_id, begin_path_time) DO UPDATE
14 SET end_path_time     = EXCLUDED.end_path_time,
15     distance_km       = EXCLUDED.distance_km,
16     max_speed_kmh     = EXCLUDED.max_speed_kmh,
17     duration_seconds = EXCLUDED.duration_seconds;

```

Listing 4.3: Idempotent upsert into fact_trip.

4.6 Analytical marts

4.6.1 Behaviour mart

The mart `marts.mart_device_monthly_behavior` carries the 35 behavioural features at the `(tenant_id, device_id, year_month)` grain and is rebuilt incrementally from the trip-related facts and the archive-derived facts. Its schema is the principal contract of the ML serving layer.

4.6.2 Telemetry mart

The mart `marts.mart_device_monthly_telemetry` is the archive-derived companion of the behaviour mart, with engine-on time, idle ratio, average and high RPM indicators, computed from the high-resolution archive table.

4.6.3 BI marts

Three additional marts feed the BI layer: `mart_fleet_daily`, `mart_vehicle_monthly` and `mart_tenant_monthly_summary`. They are exposed through dedicated views (`v_executive_dashboard`, `v_operational_dashboard`, `v_maintenance_dashboard`) consumed by the BI dashboard.

4.6.4 Unified ML view

The view `v_ml_features_full` unifies the behaviour and the telemetry marts into a single contract for the modeling notebooks; the companion view `v_device_risk_profile` computes the weighted risk score and its categorical band.

4.7 Feature engineering

4.7.1 Feature registry

The catalogue of features is declared in `config/feature_definitions.yaml`. Each feature is described by its name, its type, its unit, its formula and the fact tables it depends on. A typed Python wrapper exposes the registry to the notebooks. Table 4.3 shows a sample of the registry.

4.7.2 Aggregation strategy

The aggregation strategy follows three principles: *normalization by exposure* (most counters are divided by 100 km of distance), *operational interpretability* (each feature corresponds to an indicator that a fleet manager recognizes) and *declarative specification* (the registry is the single source of truth between the SQL transformations and the Python feature consumption). The 100 km denominator was selected because it is commonly used in fleet and road-safety reporting and makes event rates comparable across devices with different mileage.

Table 4.3: Sample of the 35-feature registry (selected groups).

Group	Feature	Definition
Volume	total_trips	Count of trips in month
Volume	total_distance_km	Sum of distances
Volume	stddev_trip_distance	Std. dev. of trip distances
Speed	avg_max_speed_kmh	Mean of per-trip max speed
Speed	p95_max_speed	95 th percentile of max speed
Speed	high_speed_trip_ratio	Trips with max speed > 100
Overspeed	overspeed_per_100km	Events / total_distance × 100
Harsh	harsh_brake_per_100km	Brake events / 100 km
Harsh	harsh_accel_per_100km	Acceleration events / 100 km
Temporal	night_trip_ratio	Trips between 20:00–06:00
Temporal	rush_hour_trip_ratio	Trips during peak hours
Telemetry	monthly_idle_ratio	Idle / engine-on minutes
Telemetry	avg_rpm	Mean RPM across day

4.7.3 Transformations

The categorical features (vehicle class, manufacturer family) are encoded with a one-hot scheme; the ordinal features (vehicle class) are encoded with an explicit ordinal scheme; the quantitative features used by the unsupervised models are standardized per tenant by means of `StandardScaler`. SMOTE was only considered in an exploratory supervised track and was not retained in the final unsupervised modeling pipeline.

4.8 Validation of the prepared dataset

4.8.1 Validation suite design

The validation suite is implemented in `sql/99_validation_suite.sql` and gathers eight check categories V1–V8: dimension population (V1), referential integrity (V2), enforcement of C1 (V3), enforcement of C2–C3 (V4), enforcement of C5 (V5), null-rate budgets on critical columns (V6), mart grain coverage (V7) and risk score distribution sanity (V8). V8 only checks the numerical validity (bounds, monotonicity, absence of NaNs) of the risk-profile view used as input by the modeling and serving layers; the final interpretation of the score is discussed in Chapter 5.

4.8.2 Final dataset summary

At the conclusion of the data preparation phase, the modeling-ready dataset comprises approximately **2,157 device-month observations** on the period 2025-01–2026-04, after applying the threshold of 100 km of total distance, positive ignition-on telemetry, and a minimum number of trips per device-month. These thresholds remove inactive or

insufficiently observed device-months whose behavioral indicators would be statistically unstable. This total includes tenant 7486, which is recovered through the telemetry-to-trip reconstruction path before the full ML feature view is consumed by Chapter 5.

4.9 Conclusion

This chapter has described the full preparation chain, from the conversion of the MySQL dump files of Accent Tunisie into PostgreSQL-importable artefacts, through the watermark-driven ETL pipeline and the cleaning rule engine (eleven operational rules C1–C11, consolidating the thirteen data-quality issues from Chapter 3), to the dimensional modeling of the warehouse, the materialization of the analytical marts, the engineering of the feature registry and the validation of the prepared dataset. The next chapter exploits the ML-ready dataset to train and to evaluate the unsupervised models that produce the behavioural segmentation and the risk score.

Chapter 5

Modeling and Evaluation of Device Behaviour Signals

5.1 Introduction

Chapter 4 produced the modeling-ready layer of the project: a PostgreSQL analytical warehouse, a validated set of marts and the unified view `marts.v_ml_features_full`. This chapter documents the modeling and evaluation work performed on top of that layer. In CRISP-DM terms, it deliberately combines Phase 4 (*modeling*) and Phase 5 (*evaluation*), because the project did not follow a linear sequence in which all models were trained first and judged later. Each modeling choice was evaluated immediately against technical metrics, business constraints and deployment readiness before the next experiment was attempted.

The chapter is therefore organized around the actual decision process of the project. It first explains why the initial supervised driver-scoring idea was reformulated into unsupervised device-month modeling. It then presents the modeling dataset, the candidate techniques, the experimental protocol, the evaluation gates and the final decision. The main outcome is a two-part analytical layer: behavioural segmentation through K-Means clustering and risk prioritization through an anomaly-oriented score. The chapter also states the limits of the evaluation, especially the narrowness of the labelled maintenance overlap, so that the deployment chapter can expose the models with the appropriate operational caveats.

5.2 Reformulation of the Modeling Problem

5.2.1 Initial target

The original ambition of the project was to build a supervised driver-risk model using historical overspeed and alert indicators as proxy labels. This formulation appeared attractive because it would have produced a direct classification target and a familiar performance vocabulary. However, the data understanding phase showed that the available sources do not support that assumption robustly.

5.2.2 Evidence from the data understanding phase

Four observations led to the reformulation. First, driver attribution is available for only a very small part of the active fleet: the usable analytical unit is therefore the *device-month*, not the driver. Second, the archive-derived harsh-event, RPM and idling signals are concentrated after 2025-01; older rows do not carry enough information for the intended modelling task. Third, harsh-event activity and overspeed activity do not behave as interchangeable labels: at the global level, they may even pull in different directions. Fourth, the tenants observed in the data do not share the same operating regime. In particular, tenant 264 is overspeed-rich, tenant 1787 is harsh-event-rich, and tenant 7486 is telemetry-first before reconstruction, making a single behavioural scale misleading.

5.2.3 Adopted modeling objective

The revised objective is to model *behavioural structure* rather than to predict an unavailable label. The problem becomes twofold:

- identify recurrent device-month behavioural profiles that fleet managers can interpret and compare;
- rank device-months by abnormality or unsafe behaviour without pretending that this rank is a validated accident or maintenance probability.

This decision is consistent with CRISP-DM: when business success criteria cannot be mapped to a reliable target variable, the modeling phase must return to the data understanding findings instead of forcing a proxy label. The adopted strategy therefore uses unsupervised learning and evaluates it through internal validity, stability, interpretability and business-facing proxies.

5.3 Modeling Dataset and Protocol

5.3.1 Input view

All experiments are based on `marts.v_ml_features_full`, the frozen feature contract produced during data preparation. The view joins the behavioural mart and the telemetry mart at the grain (`tenant_id`, `device_id`, `year_month`). The full analytical feature registry contains thirty-five features, while the production clustering pipeline uses a compact subset of thirteen high-signal indicators: overspeed rate, speed-over-limit tendency, high-speed-trip ratio, alert rate, harsh-brake rate, harsh-acceleration rate, harsh-cornering rate, idle ratio, high-RPM intensity, night-driving ratio, rush-hour ratio,

trip-distance variability and short-trip ratio.

5.3.2 Filtering rules

The modeling notebooks apply three filters before training:

- `year_month >= '2025-01'`, because the archive-derived harsh, RPM and idling features are not sufficiently populated before that date;
- `total_distance_km >= 100`, to remove inactive or almost inactive device-months that would create a trivial cluster;
- `total_ignition_on_minutes > 0`, to guarantee that telemetry-derived ratios are meaningful.

After filtering, the evaluated tenants are 235, 238, 264, 1787 and 7486, for a total of 2,157 device-month observations in the evaluation tables. Tenant 7486 contributes 434 device-months after telemetry-to-trip reconstruction, which prevents the archive-only tenant from being silently dropped from the modeling phase. This corpus is small enough to make interpretability and stability more important than model complexity.

5.3.3 Tenant-aware fitting

The research notebooks fit one model per tenant. This choice prevents the most active tenant from dominating the geometry of the model and preserves tenant-specific operating regimes. Each tenant slice is standardized independently with `StandardScaler`; K-Means and Isolation Forest are both executed with `random_state=42`. For the API and batch-scoring path, the current v1 industrialized clustering module persists a single K-Means pipeline through MLflow or local `joblib` files; per-tenant registered models are treated as the next production hardening step. This distinction matters: the research verdict is per-tenant, while the deployed software contract currently exposes a v1 serving implementation.

5.3.4 Reproducibility

The modeling and evaluation notebooks are deterministic and can be re-run from the database without depending on intermediate parquet files. The modeling outputs are stored as `data/ml/device_clusters.parquet` and `data/ml/device_risk_scores.parquet`; the serving layer stores cluster assignments in the `marts` cluster-assignment fact table. Tenant 7486 is protected by the reconstruction script `scripts/reconstruct_telemetry_trips.py`, which now recomputes both the trip-side and telemetry-side `marts` before refreshing risk and cluster outputs. The SQL risk profile is mirrored by the

Python reference implementation in `src/accident_fleet/features/risk_score.py`, which allows the same score to be computed inside PostgreSQL and by the API.

5.4 Candidate Modeling Techniques

5.4.1 Rule and statistical baselines

Two baselines were retained to keep the evaluation grounded. The first one is a rule-based risk profile computed from explicit business factors: overspeed rate, severe overspeed share, high-speed-trip ratio, speed-alert rate, night-driving ratio and maximum-speed tendency. These factors are normalized and weighted in `config/feature_definitions.yaml`; the SQL implementation is `marts.v_device_risk_profile`. The second baseline is a Principal Component Analysis (PCA) projection used as a diagnostic view of the main axis of feature variance, not as a final operational model.

5.4.2 K-Means clustering

K-Means was selected as the principal segmentation candidate because it is simple, fast, reproducible and easy to explain to non-technical stakeholders [1]. For each tenant, the number of clusters is selected by sweeping $k \in \{3, 4, 5, 6\}$ and retaining the value that maximizes the silhouette coefficient. PCA with two components is used only for visualization; it is not the training space of the model.

5.4.3 Alternative clustering methods

Hierarchical clustering with Ward linkage was used as a comparison point. It produced interpretable partitions, but its operational cost and weaker serving simplicity made it less attractive for the production path. DBSCAN was also tested, but on this dataset it tended to produce one dominant cluster and a tail of very small groups, which is difficult to use in a fleet-management dashboard.

5.4.4 Anomaly scoring

Isolation Forest was evaluated as the unsupervised anomaly-scoring candidate [2]. The model scores how unusual a device-month is relative to its tenant cohort. The raw decision score is rescaled within each tenant to the interval $[0, 1]$ and then split into *low*, *medium* and *high* bands for evaluation. This score is best interpreted as an abnormal-behaviour indicator, not as a probability of failure.

5.4.5 Deferred candidates

One-Class SVM and Local Outlier Factor were examined but not retained. One-Class SVM is sensitive to kernel and bandwidth choices; Local Outlier Factor is sensitive to neighbourhood size and becomes unstable on smaller tenant slices. Deep-learning baselines, including an autoencoder and a proxy-label Multi-Layer Perceptron, were also deferred. The available corpus is too small and too weakly labelled to justify their complexity.

5.5 Evaluation Design

5.5.1 CRISP-DM evaluation gates

The evaluation phase checks whether the models are useful for the business problem, not only whether they produce acceptable technical metrics. The gates are:

- *cluster validity*: silhouette at least 0.15 per tenant, supported by Davies–Bouldin and Calinski–Harabasz metrics [3, 4, 5];
- *cluster usefulness*: at least three profiles and no unexplained persona labels;
- *risk consistency*: high-risk rows must show higher unsafe-feature means than low-risk rows;
- *temporal stability*: cluster churn and risk-band churn must remain below their operational gates;
- *coverage and fairness*: no tenant should collapse into a single risk band.

5.5.2 External validation limitation

The preferred validation would compare risk scores with accident, claim or maintenance outcomes. This validation remains limited on the current modeled population. The `staging.sinistre` table is empty, the `staging.offense` table contains too few rows to support inference, and the available maintenance events overlap the modeling frame essentially through tenant 7486 only. Tenant 7486 is now present in `marts.v_ml_features_full`, but a single-tenant maintenance overlap cannot validate the score for all fleets. The maintenance-based validation is therefore treated as tenant-limited directional evidence, not as a full outcome calibration.

5.5.3 Fallback validation

Because external labels are unavailable, the evaluation relies on internal consistency and stability. Internal consistency asks whether the high-risk band is actually higher than the low-risk band on unsafe behavioural features. Stability asks whether a device remains in a comparable cluster or band from one month to the next unless its behaviour changes. These tests do not replace outcome validation, but they are legitimate CRISP-DM Phase 5 evidence for deciding whether the artefact can be deployed as a decision-support indicator with caveats.

5.6 Experimental Results

5.6.1 Cluster quality

Table 5.1 reports the deterministic K-Means evaluation. All tenants meet the silhouette gate of 0.15 and the minimum k gate of three clusters.

Table 5.1: Per-tenant K-Means quality metrics.

Tenant	n	k	Sil.	DB	CH
235	613	6	0.225	1.271	110.5
238	339	6	0.236	1.165	72.2
264	354	6	0.174	1.627	52.9
1787	417	5	0.269	1.313	82.5
7486	434	5	0.253	1.374	61.9

The result is now interpreted on five tenants. All tenants pass the silhouette and minimum-cluster gates, and none has a cluster above the 70% dominance threshold. The tenant 7486 solution contains one large normal cluster covering 65.4% of rows and several very small exceptional clusters, which is acceptable for analysis but should be labelled carefully in the dashboard.

5.6.2 Risk-score internal consistency

Table 5.2 summarizes the internal-consistency fallback. For each tenant, seven unsafe-direction features were compared between the high and low risk bands. Every tenant clears the gate: at least five features have a high-band mean greater than the low-band mean, and at least two features have a ratio of two or more.

For example, in tenant 7486 the high band has approximately 4.5 times more overspeed events, 5.4 times more harsh braking, 5.5 times more harsh acceleration, 10.5 times more harsh cornering and 18.5 times more high-RPM minutes than the low band. This

Table 5.2: Internal consistency of the anomaly risk score.

Tenant	Ratio > 1	Ratio $\geq$ 2	Mean ratio (approx.)
235	6/7	5	20.25
238	5/7	3	21.98
264	5/7	4	20.24
1787	6/7	6	11.90
7486	7/7	6	6.84

supports the interpretation of the score as a ranking of abnormal and unsafe behaviour, while still falling short of proving a causal relationship with future maintenance events.

5.6.3 Temporal stability

Table 5.3 reports the month-to-month stability results available from the committed stability notebook. That run predates the tenant 7486 reconstruction and is therefore retained as historical evidence for the original four tenants, not as the final five-tenant release gate. The code path now protects tenant 7486 from being dropped, and the next stability rerun must include it before production promotion. On the executed four-tenant run, cluster churn remains between 5.5% and 14.0%; risk-band churn remains between 10.7% and 15.3%, both well below the notebook gates.

Table 5.3: Month-to-month stability by tenant.

Tenant	Month pairs	Cluster churn	Band churn
235	507	0.055	0.116
238	279	0.140	0.108
264	295	0.075	0.153
1787	326	0.129	0.107
7486	Pending five-tenant stability rerun after reconstruction		

The historical risk-band transition matrix, shown in Table 5.4, is also diagonal-heavy. A device-month in the low band remains low in the following month with probability 0.935, while a high-band row remains high with probability 0.600. This confirms that the score is not random from one month to the next on the four-tenant stability run; tenant 7486 must be included in the next execution of the same gate.

Table 5.4: Risk-band transition matrix between consecutive months.

$T / T + 1$	Low	Medium	High
Low	0.935	0.063	0.002
Medium	0.292	0.639	0.069
High	0.050	0.350	0.600

5.7 Model Decision and Final Justification

5.7.1 Decision table

Table 5.5 summarizes the final decision for each artifact.

Table 5.5: Final modeling and evaluation decision.

Artifact	Decision	Justification
K-Means clusters	Ship for the five modeled tenants, with careful labels for tiny exceptional clusters.	Tenants 235, 238, 264, 1787 and 7486 pass the silhouette, minimum- k and dominance gates.
Isolation-Forest risk band	Ship provisionally as an anomaly indicator.	Internal consistency passes for all five tenants; outcome backtesting is still tenant-limited and not a full calibration.
Composite SQL/Python risk profile	Keep as the deployed scoring contract for API and dashboard risk lookup.	The score is transparent, auditable and identical in SQL and Python, which supports operational explainability.
Deep-learning candidates	Defer.	The corpus is too small and lacks labels, making deep models harder to justify and harder to explain.

5.7.2 Why the final solution is mixed

The final solution is not a single black-box model. It is a controlled combination of research-grade unsupervised modelling and production-grade scoring contracts. K-Means provides behavioural segmentation. Isolation Forest provides an anomaly-oriented research score that is useful for ranking and validation. The SQL/Python composite risk profile provides the explainable score used by the serving layer. This mixed design reflects the maturity of the available data: enough structure exists to segment and prioritize devices, but not enough labelled evidence exists to claim supervised prediction of incidents.

5.7.3 Operational caveat

The following caveat must accompany the risk score in the dashboard and in any business presentation:

Risk scores were validated against feature-space anomalies. Outcome backtesting is currently tenant-limited and is pending integration of a broader incident or claims feed.

This formulation protects the project from overclaiming. It also provides a concrete next action for the host organization: integrate a labelled incident, claim or maintenance feed that overlaps with all modeled tenants, not only tenant 7486.

5.8 Model Outputs for Deployment

The deployment phase receives three model-related contracts. First, the clustering parquet export contains exploratory cluster labels at the (`tenant`, `device`, `month`) grain. Second, the risk-score parquet export contains the anomaly score used by the evaluation notebooks. Third, the industrialized pipeline writes cluster assignments into the marts cluster-assignment fact table and exposes the transparent risk profile through a materialized risk-profile table. These contracts allow the dashboard to display behavioural groups, risk categories and lineage information without recomputing models at page-load time.

Model monitoring is defined as part of the deployment hand-off. Every monthly extension of the modeling window must re-run the evaluation notebooks, refresh the cluster and risk-score artifacts, and verify that no gate has regressed. The drift module computes Population Stability Index on the feature distributions, while the promotion module prevents a clustering candidate from replacing the current model when the silhouette drop exceeds the configured tolerance.

5.9 Conclusion

This chapter has regenerated the modeling phase as a combined CRISP-DM modeling and evaluation chapter. The initial supervised driver-scoring objective was rejected because the data does not provide reliable driver labels or outcome labels for the modeled tenants. The final analytical strategy therefore uses device-month unsupervised modeling, tenant-aware evaluation and transparent risk scoring across tenants 235, 238, 264, 1787 and 7486. K-Means is retained for behavioural segmentation, Isolation Forest is retained as a provisional anomaly indicator, and the deployed SQL/Python score is kept as the explainable operational contract. The next chapter presents how these artefacts are exposed through the Azure-hosted pipeline, the API, the dashboard and the monitoring layer.

Chapter 6

Deployment, API, Dashboard and Azure Operations

6.1 Introduction

The evaluation of Chapter 5 has confirmed the readiness of the behavioural segmentation and of the transparent risk profile for controlled industrialization, while keeping the Isolation Forest anomaly score as a caveated evaluation artefact. The present chapter executes the deployment phase of CRISP-DM. The chapter describes the deployment strategy, the API and the dashboard exposing the analytical outputs, the integration with the analytical PostgreSQL warehouse populated from the converted MySQL dumps of Accent Tunisie, the cron-driven automation, the monitoring and alerting layer, and finally the Microsoft Azure infrastructure on which the platform runs, with a particular focus on the secure remote collaboration setup that allows the two-student team of engineers to operate the platform from separate workstations through SSH tunnelling, IP allowlisting on the Azure Network Security Group, and dedicated service accounts.

6.2 Deployment strategy

6.2.1 Industrialization principles

Three principles guide the industrialization. The first one is the *freshness* of the analytical outputs, with a target latency of less than ten minutes between the appearance of an event in the source platform and its visibility in the marts. The second one is the *idempotency* of every step, which allows safe replays. The third one is the *observability*, which guarantees that any incident is detected and diagnosed without manual inspection.

6.2.2 Operating modes

The pipeline supports three operating modes accessible through a single command-line entry point (`scripts/run_batch.py`):

- `-mode bootstrap`: one-off DDL and full-refresh of the dimensions and state tables.
- `-mode backfill`: chunked historical replay of the staging tables (default chunk: 30 days).
- `-mode incremental`: default operating mode, processing the slice strictly after the watermark, scheduled every five minutes.

6.3 Microsoft Azure infrastructure

6.3.1 Azure Virtual Machine

The platform is hosted on a Microsoft Azure Linux virtual machine. The VM hosts, on the same operating system, the analytical PostgreSQL instance, the Python virtual environment, the Prefect flow, the API process, the BI dashboard process, the cron entry and the system journal. The git repository of the project is cloned into `/opt/accent-fleet-analytics` and a dedicated Linux user (`azureuser`) owns the runtime artefacts.

6.3.2 Network Security Group and IP allowlist

The VM is associated with an Azure Network Security Group that closes every public port except SSH (port 22) and the HTTPS port of the reverse proxy (port 443). The SSH port is itself restricted to an explicit allowlist of source IP addresses corresponding to the personal workstations of the two-student team and to the office network of Accent Tunisie. This combination ensures that no PostgreSQL port, no API port and no dashboard port is exposed to the public internet, while still allowing both members of the team to connect to the shared analytical environment from their respective home workstations.

6.3.3 Secure remote collaboration via SSH tunnelling

Day-to-day collaboration of the team relies on SSH tunnelling. Each team member opens a local tunnel that forwards the local PostgreSQL port to the remote port on the VM, as illustrated in Listing 6.1. The notebooks and the local clients then connect to `localhost:5432` as if the database were running locally, while the actual traffic is encrypted by SSH.

```
1 # Forward local 5432 -> remote 5432 over SSH
2 ssh -i ~/.ssh/azure_id_ed25519 \
3     -L 5432:localhost:5432 \
```

```
4 -N \
5 azureuser@<vm-public-ip>
```

Listing 6.1: Opening an SSH tunnel from a developer workstation to the Azure VM.

6.3.4 Service accounts

Three categories of accounts coexist on the VM: *personal accounts*, one per student, used only through SSH key authentication and intended exclusively for human interactive sessions; an *automation account* (`azureuser`) used by cron and by the systemd services; and a *database service account* (`accent_pipeline`) used by the pipeline to authenticate against PostgreSQL, with credentials externalized in an environment file. Each student uses their own SSH key pair, which means that an audit of the system journal can attribute every interactive command to its actual operator without ambiguity.

6.4 PostgreSQL integration

6.4.1 Analytical PostgreSQL on the VM

The analytical PostgreSQL is provisioned in version 16 on the VM, with the three schemas `staging`, `warehouse` and `marts`, plus the operational tables `etl_watermark`, `etl_run_log` and `quarantine_rejected`. It is populated by the conversion bridge of Chapter 4, which loads the converted MySQL dumps of Accent Tunisie into the staging schema, and then by the watermark-driven pipeline described in this chapter. The instance listens only on `localhost`, which is why remote access from the two students' workstations is mediated by the SSH tunnel described above.

6.4.2 Connection management

The pipeline manages the database connection through a `SQLAlchemy` engine with a connection pool, an automatic ping before each transaction (`pool_pre_ping`) and an explicit timeout. The credentials are read from a `.env` file owned by `azureuser`, with restrictive file permissions.

6.4.3 Backups and snapshots

A nightly logical backup of the analytical database is exported through `pg_dump` and rotated through a seven-day retention. A weekly full snapshot of the disk is configured at the Azure infrastructure level and is retained for four weeks.

6.5 API and backend

6.5.1 API design

The API is implemented with FastAPI [6] and exposes a small number of endpoints, summarized in Table 6.1. Authentication is handled through API keys associated with the consumer applications.

Table 6.1: API endpoints exposed by the platform.

Method	Endpoint	Description
GET	/health	Liveness probe.
GET	/devices/{id}/risk	Latest risk score and band of a device.
GET	/devices/{id}/cluster	Latest cluster assignment of a device.
GET	/tenants/{id}/leaderboard	Top-N high-risk devices of a tenant.
GET	/tenants/{id}/kpi	Tenant-month KPI bundle.
POST	/admin/refresh	Force an off-cycle incremental run.

6.5.2 Backend architecture

The API is served by Uvicorn behind an Nginx reverse proxy bound to 127.0.0.1. A systemd unit ensures the API restarts automatically after a crash or after a VM reboot. The API queries the marts and views directly through `psycpg2`, with prepared statements and a small connection pool.

6.5.3 Reverse proxy and TLS

The Nginx reverse proxy listens on port 443 and terminates TLS with a certificate provisioned through Let's Encrypt. It forwards the HTTPS traffic to the local Uvicorn process. Access logs are written to the system journal under a dedicated tag.

6.6 Dashboard and frontend

6.6.1 Dashboard implementation

The dashboard is implemented with Streamlit (or, equivalently, Plotly Dash) and is served as a systemd unit, behind the same Nginx reverse proxy as the API. The dashboard queries the BI views (`v_executive_dashboard`, `v_operational_dashboard`, `v_maintenance_dashboard`, `v_fleet_risk_dashboard`) through a read-only database connection.

6.6.2 Pages and visualizations

The dashboard is organized in five pages: the executive overview (tenant-month KPIs and rolling deltas), the operational view (daily KPIs and seven-day rolling), the maintenance leaderboard, the risk heatmap (cluster $\times$ band $\times$ tenant) and the device-detail page (history of the risk score, cluster trajectory, alerts). Figure 6.1 sketches the architecture of the dashboard.

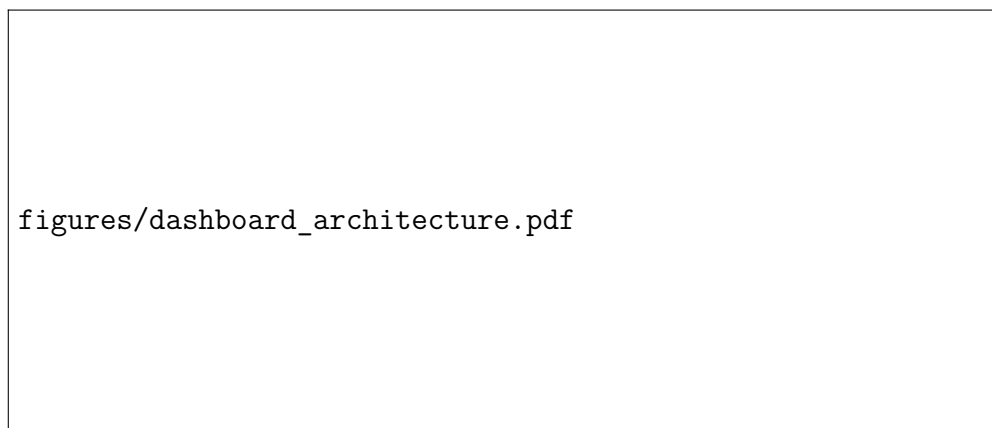


Figure 6.1: Architecture of the BI dashboard.

6.7 Automation

6.7.1 Cron entry

The incremental mode is triggered every five minutes by a cron entry installed under `/etc/cron.d/accent-fleet`. The entry calls the pipeline through the Python virtual environment of the project and pipes the standard output and the standard error to the system journal under the tag `accent-fleet`, as shown in Listing 6.2.

```

1 */5 * * * * azureuser cd /opt/accent-fleet-analytics && \
2   /opt/accent-fleet-analytics/.venv/bin/python scripts/
   run_batch.py \
3   --mode incremental 2>&1 | logger -t accent-fleet

```

Listing 6.2: Cron entry triggering the incremental pipeline every five minutes.

6.7.2 Prefect flow

Internally, the pipeline is implemented as a Prefect flow defined in `src/accent_fleet/pipeline/flow_batch.py`. The flow chains the extraction tasks, the cleaning rule engine, the loading of dimensions and facts, the materialization of the marts, the

validation suite and the export of the model artefacts. Each task is idempotent and emits structured logs.

6.7.3 Backfill orchestration

The backfill mode reuses the same Prefect flow but iterates over fixed-size chunks (default of 30 days) of the staging history. The mode is invoked manually by an operator after a schema change, after a model retraining or after a substantial cleaning-rule revision.

6.7.4 Telemetry reconstruction for tenant 7486

Tenant 7486 is present in the archive telemetry before it is present in the original trip table. The deployment therefore includes the reconstruction command `scripts/reconstruct_telemetry_trips.py`, which rebuilds trip-like records from ignition-on telemetry, recomputes the trip-side and telemetry-side marts, refreshes the materialized risk profile and attempts cluster scoring for the touched months. This step prevents tenant 7486 from disappearing from the modeling and dashboard views after a fresh backfill.

6.8 Monitoring and alerting

6.8.1 Run log and lineage

Every pipeline execution writes a row in `warehouse.etl_run_log`, with the run identifier, the operating mode, the start and end timestamps, the status and the per-fact processed row counts. This table is the principal observability surface of the platform.

6.8.2 Data quality monitoring

The validation suite (V1–V8) is executed at the end of every cycle, and its quantitative outcomes are persisted in a dedicated table. A time series of the data quality measurements is plotted in the executive dashboard.

6.8.3 Model drift monitoring

Model drift is monitored by tracking, on a rolling basis, the proportions of device-months in each K-Means cluster and in each risk band. A statistically significant deviation triggers an alert and motivates a retraining.

6.8.4 Alerting

The alerting layer is implemented in `src/accnet_fleet/monitoring/` and watches three signals: pipeline failures (status of the run log), data-quality regressions (validation suite) and model drift. When a threshold is crossed, a notification is sent to the operations channel (Slack webhook or e-mail).

6.8.5 Operations runbook

A short runbook documents the manual procedures: how to inspect the run log, how to trigger a backfill, how to restart the API or the dashboard, how to rotate the database credentials and how to renew the TLS certificate. The runbook is reproduced in Appendix C.

6.9 Production deployment workflow

6.9.1 Code deployment

The deployment of new code follows a simple workflow: a pull request is reviewed and merged on the `main` branch of the git repository; the operator pulls the new commits into `/opt/accnet-fleet-analytics` on the VM; the dependencies are upgraded if needed; the API and the dashboard systemd units are restarted; and the next cron cycle exercises the new pipeline.

6.9.2 Database migrations

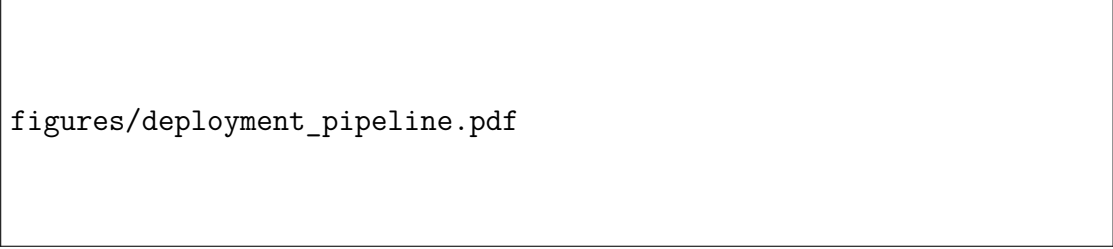
DDL changes follow a versioned migration pattern. Every change is shipped as a numbered SQL file under `sql/` and is applied by the bootstrap mode of the pipeline (or by a manual `psql` invocation under operator control). The migration scripts are idempotent: a re-execution does not corrupt the schema.

6.9.3 Model deployment

Model deployment relies on the artefact store described in Chapter 5. A new clustering candidate is logged with its scaler, feature order and metrics, then promoted only if the evaluation gate accepts it. The API loads the promoted model from the registry when available, or from the local `models/clustering/` fallback directory, and can reload the artefact without changing the database contract.

6.9.4 Diagram of the deployment pipeline

The deployment pipeline is summarized in Figure 6.2.



figures/deployment_pipeline.pdf

Figure 6.2: End-to-end deployment pipeline of the platform.

6.10 Conclusion

This chapter has formalized the deployment of the data engineering and machine learning platform on the Microsoft Azure infrastructure, with explicit attention to the analytical PostgreSQL warehouse fed from the converted MySQL dumps of Accent Tunisie, to the secure remote collaboration setup that allows the two-student team of engineers to operate the platform from separate workstations (SSH tunnelling, IP allowlist on the Network Security Group, per-student service accounts), to the API and the BI dashboard, to the cron-driven automation, and to the layered monitoring and alerting plan. The end-to-end CRISP-DM cycle is therefore brought to its operational completion. The general conclusion that follows synthesizes the deliverables, the lessons learned and the perspectives opened by the project.

General Conclusion

The work documented in this thesis has covered the full CRISP-DM cycle of an industrial data engineering and machine learning project conducted at Accent Tunisie, on a multi-tenant telematics fleet of approximately six hundred GPS devices spread across five analytical tenants, including the telemetry-first tenant 7486. The starting point was an operational ecosystem composed of three MySQL databases, namely a raw hexadecimal telemetry stream populated by SIM-driven device frames, a decoded archive of `arch_XXXXX` tables retaining about six months of readable data, and a `user_XXXXX` master-data database carrying customers, vehicles and drivers, on top of which the company already exposed a set of descriptive dashboards but no advanced analytical layer. The endpoint is a production-grade analytical platform deployed on a Microsoft Azure virtual machine, refreshed every five minutes, equipped with an explicit MySQL-to-PostgreSQL conversion bridge, a cleaning rule engine, a dimensional warehouse, a registry of thirty-five behavioural features, a tenant-aware K-Means segmentation, an evaluated anomaly risk indicator, a transparent deployed risk profile, an API and a business intelligence dashboard.

The first half of the project has invested heavily in the data engineering layer. A dedicated bridge converts the `.sql` dump files received from Accent Tunisie, covering the slices of the `arch_XXXXX` archive and the `user_XXXXX` master data relevant to the analytical scope, into a PostgreSQL-compatible representation. Eleven data quality issues were identified during the data understanding phase and were formalized as cleaning rules executed at the warehouse boundary. A dimensional model with five dimensions, a bridge and eleven facts has stabilized the silver layer, on top of which the analytical marts and the unified ML view have been built. The watermark-driven incremental pipeline, hosted on the Azure VM and orchestrated by Prefect, has reduced the per-cycle processing time to a few seconds while preserving idempotency and replayability, and has done so without ever touching the operational MySQL workload of the company.

The second half of the project has built the analytical and serving layers. After a comparative study covering rule-based, statistical, classical machine-learning and deep-learning candidates, the retained solution combines K-Means behavioural segmentation, an Isolation Forest anomaly indicator evaluated with explicit caveats, and a transparent SQL/Python risk profile for serving. The behavioural profiles obtained align with the operational vocabulary of the fleet managers; the risk categories feed the business intelligence dashboard and the API consumed by the consumer applications. A layered

monitoring covers the pipeline health, the data quality and the model drift, with alerting hooked to the operations channel. Together, these deliverables move the analytical centre of gravity of Accent Tunisie from a purely descriptive surface, answering the question *what happened?*, to a forward-looking decision-support layer that answers *which devices require attention next month?*

A particular care has been given to the secure remote collaboration on the Azure infrastructure, dictated by the two-student nature of the project. Both members of the team worked from separate personal workstations and shared the analytical environment through their respective SSH key pairs. The PostgreSQL ports are not exposed to the public internet; access is mediated by SSH tunnels, themselves protected by a strict IP allowlist applied at the Azure Network Security Group level and limited to the two students' workstations and the office network of Accent Tunisie. Per-member personal accounts coexist with a dedicated automation account on the VM, under controlled file permissions; the database service account is provisioned with the minimal privileges required by the pipeline.

Three lessons emerge from the project. *First*, the early formalization of the cleaning rules has been the single most effective decision: it has prevented the propagation of source-side anomalies into the modeling layer and has made every transformation auditable, an essential property when the source is an external dump file rather than a directly accessible database. *Second*, the per-tenant fitting protocol has been a decisive factor in obtaining operationally meaningful clusters; a global model would have been dominated by the most active tenant and would have failed to capture the heterogeneity of the regimes anticipated in the data understanding phase. *Third*, the strict separation between the operational MySQL stack of Accent Tunisie and the analytical PostgreSQL warehouse of the project has been more than an administrative convenience: it has freed the analytical work to iterate on cleaning rules, feature definitions and modeling choices without ever putting the customer-facing platform of the company at risk.

Three perspectives are open for future iterations. The first one is the integration of a labelled incident feed from the operations team of Accent Tunisie, which would unlock a supervised classification track and a quantitative calibration of the risk score against ground truth. The second one is the migration of the dump-based ingestion towards a streaming back-end (Kafka, Redpanda or a managed equivalent), in order to bring the latency below the minute and to support event-driven downstream consumers; this would naturally complement the existing streaming-style ingestion of the raw hexadecimal telemetry on the operational side. The third one is the introduction of a hierarchical or multi-task model capable of leveraging the structural similarities across tenants while preserving the tenant-specific regimes already identified.

Beyond the technical deliverables, the project has been a formative experience on the trade-offs between elegance and operational sustainability, and between methodological rigour and the very concrete constraints of a two-student collaboration. Building a pipeline that runs successfully once is comparatively easy; building a pipeline that runs every five minutes for several months without manual intervention, on an infrastructure jointly operated by two student engineers from separate workstations, requires a different kind of engineering, namely one in which observability, idempotency, explicit failure modes and disciplined access control carry as much weight as algorithmic sophistication. The platform delivered in this thesis aims to incarnate that discipline.

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Appendix A

SQL Exploration Scripts

This appendix references the SQL scripts used during the data understanding and the data preparation phases. The full versions are stored in the project repository under `scripts/sql/`.

A.1 Schema inspection

The script `01_schema_inspection.sql` returns the column-level metadata of every table in the staging and warehouse schemas, by querying `information_schema.columns`, `information_schema.table_constraints` and `information_schema.key_column_usage`. The result feeds the data dictionary and the schema map.

A.2 Data profiling

The script `02_data_profiling.sql` computes, for every column, descriptive statistics: row count, distinct count, null count, minimum, maximum, mean, standard deviation, and the 5th, 50th, 95th percentiles. The script is parametrized by the table and the column name and is invoked through a shell wrapper that iterates over the catalogue.

A.3 Null analysis

The script `03_null_analysis.sql` produces the null rates of every column of the principal source tables, in a format directly consumable by the missing-value heatmap rendered by `fig_missing_heatmap.py`.

A.4 Distributions

The script `04_distributions.sql` computes per-column histograms with a configurable number of bins, using the `width_bucket` function of PostgreSQL. The result feeds the distribution plots produced in Chapter 3.

A.5 KPI extraction

The script `05_kpi_extraction.sql` computes the principal business KPIs at the tenant-month grain (total trips, total distance, average max speed, overspeed-per-100km, harsh-event-per-100km, idle ratio) directly from the marts. The result is consumed by the executive dashboard and by the figures of Chapter 5.

A.6 Metadata analysis

The script `06_metadata_analysis.sql` returns the size, the row count and the last analyse-timestamp of every table, by querying `pg_class`, `pg_stat_user_tables` and `pg_stat_all_indexes`. The script is used during the operational reviews of the platform.

A.7 Validation suite

The validation suite `99_validation_suite.sql` aggregates the eight checks V1–V8 described in Chapter 4 and writes a synthetic report into a dedicated table consumed by the monitoring layer.

Appendix B

Python Scripts and Figure Generation

This appendix references the Python scripts used in the project. The full versions are stored under `scripts/python/`.

B.1 Automated extraction to CSV

The script `extract_to_csv.py` reads a manifest of SQL queries from `config/queries.yaml`, executes each query through a SQLAlchemy engine and writes the result as a time-stamped CSV in `exports/`. The script accepts a `-dry-run` flag for unit-testing purposes.

B.2 Figure generation

The figures of the thesis are produced by self-contained Python scripts located in `scripts/python/figures/`. Each script reads its input either from the database (read-only credentials) or from a CSV produced by `extract_to_csv.py`, and writes a publication-quality PDF in `report/figures/`. The scripts and the figures they produce are listed in Table B.1.

Table B.1: Figure-generation scripts referenced in the thesis.

Script	Figure produced
<code>fig_crispdm_workflow.py</code>	Diagram of the CRISP-DM workflow.
<code>fig_architecture.py</code>	Logical architecture of the platform.
<code>fig_schema_map.py</code>	Schema map of the source tables.
<code>fig_missing_heatmap.py</code>	Missing-value heatmap.
<code>fig_distributions.py</code>	Distribution plots of the principal variables.
<code>fig_feature_importance.py</code>	Surrogate feature importance.
<code>fig_model_comparison.py</code>	Comparative dashboard of the candidate models.
<code>fig_confusion_matrix.py</code>	Confusion matrix between rule and Isolation Forest.
<code>fig_api_architecture.py</code>	Architecture of the API.
<code>fig_deployment_pipeline.py</code>	End-to-end deployment pipeline.
<code>fig_azure_topology.py</code>	Azure VM topology with NSG and SSH tunnel.

B.3 Modeling scripts

The modeling notebooks are stored under `notebooks/04_modeling/`. The notebook `01_device_behavior_clustering.ipynb` fits the per-tenant K-Means models and exports the artefacts; the notebook `02_anomaly_risk_score.ipynb` fits the per-tenant Isolation Forest models and computes the rescaled risk scores.

Appendix C

Azure Operations Runbook

This appendix collects the operational procedures of the Microsoft Azure environment hosting the platform. It is intended as a quick reference for the operations team of Accent Tunisie.

C.1 VM provisioning

C.1.1 VM specification

The platform runs on an Azure Linux virtual machine sized for a moderate analytical workload. The VM is provisioned in a Tunisian-or-European region of the Azure cloud, with a managed OS disk and an additional managed data disk dedicated to the PostgreSQL data directory. A static public IP address is associated with the VM.

C.1.2 Network Security Group

The Network Security Group (NSG) attached to the VM applies the following rules:

- **Inbound 22/tcp** (SSH): allowed only from the IP allowlist of the team and of the office network of Accent Tunisie.
- **Inbound 443/tcp** (HTTPS): allowed for the API and the dashboard, behind Nginx with a Let's Encrypt certificate.
- **Inbound 80/tcp** (HTTP): allowed only for the ACME challenge of Let's Encrypt; redirected to HTTPS otherwise.
- **Inbound 5432/tcp** (PostgreSQL): denied. PostgreSQL listens only on `localhost` and is reachable through SSH tunnels.

C.1.3 IP allowlist management

The IP allowlist is maintained as an explicit `ipset` attached to the NSG. Adding or removing a member of the team is a privileged operation that requires the approval of the platform owner. The history of the allowlist changes is preserved in the operations journal.

C.2 SSH access

C.2.1 Account types

Three account categories share the VM: personal accounts (one per team member, with SSH key authentication only and no shell password), an automation account (`azureuser`) used by cron and by the systemd services, and the database service account (`accent_pipeline`) used internally by the pipeline.

C.2.2 SSH tunnel

The SSH tunnel for PostgreSQL is opened by the team members on their workstation as shown in Listing C.1. Once the tunnel is open, a notebook or a database client connecting to `localhost:5432` reaches the analytical PostgreSQL on the VM.

```
1 ssh -i ~/.ssh/azure_id_ed25519 \  
2     -L 5432:localhost:5432 \  
3     -N \  
4     azureuser@<vm-public-ip>
```

Listing C.1: SSH tunnel for PostgreSQL access.

C.2.3 Key rotation

The SSH keys are rotated annually or upon departure of a team member, whichever comes first. The rotation procedure is recorded in the operations journal.

C.3 Pipeline operations

C.3.1 Cron entry

The cron entry under `/etc/cron.d/accent-fleet` triggers the incremental pipeline every five minutes (cf. Listing 6.2 of Chapter 6). The output of the run is piped to the system journal under the tag `accent-fleet`.

C.3.2 Inspecting the run log

The history of the pipeline runs is observed through the SQL query of Listing C.2.

```
1 SELECT run_id, mode, status, started_at, ended_at,  
       rows_processed, error_message  
2 FROM warehouse.etl_run_log
```

```
3 ORDER BY started_at DESC
4 LIMIT 20;
```

Listing C.2: Inspecting the run log.

C.3.3 Triggering a backfill

A backfill is triggered manually under operator control by Listing C.3. The backfill runs in a `tmux` session so that the operator can disconnect without interrupting the run.

```
1 tmux new -s backfill
2 cd /opt/accent-fleet-analytics
3 .venv/bin/python scripts/run_batch.py --mode backfill --chunk-
  days 30
4 # Detach with Ctrl-b, d
```

Listing C.3: Triggering a chunked backfill.

C.4 Database operations

C.4.1 Backup

A nightly logical backup is exported through `pg_dump` to `/opt/backups/`, rotated through a seven-day retention policy. A weekly full snapshot of the data disk is configured at the Azure infrastructure level and retained for four weeks.

C.4.2 Credential rotation

The credentials of the service account `accent_pipeline` are rotated quarterly. The rotation procedure updates the password in PostgreSQL, in the `.env` file of the VM, and restarts the API and the dashboard systemd units to reload the connection pool.

C.5 API and dashboard operations

C.5.1 Service management

The API and the dashboard are managed as systemd units `accent-api.service` and `accent-dashboard.service`. Listing C.4 shows the canonical commands to inspect, restart and follow the logs of these services.

```
1 sudo systemctl status accent-api.service
2 sudo systemctl restart accent-api.service
```

```
3 sudo journalctl -u accent-api.service -f
4
5 sudo systemctl status accent-dashboard.service
6 sudo systemctl restart accent-dashboard.service
7 sudo journalctl -u accent-dashboard.service -f
```

Listing C.4: Managing the API and dashboard systemd units.

C.5.2 TLS certificate renewal

The TLS certificate of Nginx is provisioned by Let's Encrypt and is renewed automatically by a daily timer. The renewal status is verified through `certbot certificates`.

C.6 Monitoring and alerting

C.6.1 Pipeline alerts

A failure of the pipeline (status `ERROR` in `etl_run_log`) triggers a notification on the operations channel. The notification carries the run identifier, the operating mode and the error message.

C.6.2 Data quality alerts

A failure of any critical check of the validation suite triggers a notification, with a link to the corresponding row of the validation table.

C.6.3 Drift alerts

A statistically significant deviation of the cluster proportions or of the risk band proportions, computed against a one-month baseline, triggers a notification accompanied by a link to the dashboard tile that visualizes the drift.